

| Status   | Condition                      | Action                                                                                               |
|----------|--------------------------------|------------------------------------------------------------------------------------------------------|
| Activate | 0x0051 to ManufacturerAccess() | Outputs <i>SafetyStatus()</i> flags on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

|      |      |      |          |      |      |      |      |       |      |       |      |      |     |      |     |
|------|------|------|----------|------|------|------|------|-------|------|-------|------|------|-----|------|-----|
| 31   | 30   | 29   | 28       | 27   | 26   | 25   | 24   | 23    | 22   | 21    | 20   | 19   | 18  | 17   | 16  |
| RSVD | RSVD | RSVD | RSVD     | RSVD | RSVD | RSVD | RSVD | RSVD  | RSVD | RSVD  | OC   | RSVD | CTO | RSVD | PTO |
| 15   | 14   | 13   | 12       | 11   | 10   | 9    | 8    | 7     | 6    | 5     | 4    | 3    | 2   | 1    | 0   |
| RSVD | OCDL | OTF  | AFE_OVRD | UTD  | UTC  | OTD  | OTC  | ASCDL | ASCD | AOLDL | AOLD | OCD  | OCC | COV  | CUV |

**RSVD (Bits 31–21):** Reserved

**OC (Bit 20):** Overcharge

1 = Detected

0 = Not Detected

**RSVD (Bit 19):** Reserved

**CTO (Bit 18):** Charge Timeout

1 = Detected

0 = Not Detected

**RSVD (Bit 17):** Reserved

**PTO (Bit 16):** Precharge Timeout

1 = Detected

0 = Not Detected

**RSVD (Bits 15):** Reserved

**OCDL (Bit 14):** Overcurrent During Discharge Latch

1 = Detected

0 = Not Detected

**OTF (Bit 13):** Overtemperature Fault

1 = Detected

0 = Not Detected

**AFE\_OVRD (Bit 12):** AFE Alert

1 = Detected

0 = Not Detected

**UTD (Bit 11):** Undertemperature During Discharge

1 = Detected

0 = Not Detected

**UTC (Bit 10):** Undertemperature During Charge

1 = Detected

0 = Not Detected

**OTD (Bit 9):** Overtemperature During Discharge

1 = Detected

0 = Not Detected

**OTC (Bit 8):** Overtemperature During Charge

1 = Detected

0 = Not Detected

**ASCDL (Bit 7):** Short Circuit During Discharge Latch

1 = Detected  
0 = Not Detected

**ASCD (Bit 6):** Short Circuit During Discharge

1 = Detected  
0 = Not Detected

**AOLDL (Bit 5):** Overload During Discharge Latch

1 = Detected  
0 = Not Detected

**AOLD (Bit 4):** Overload During Discharge

1 = Detected  
0 = Not Detected

**OCD (Bit 3):** Overcurrent During Discharge

1 = Detected  
0 = Not Detected

**OCC (Bit 2):** Overcurrent During Charge

1 = Detected  
0 = Not Detected

**COV (Bit 1):** Cell Overvoltage

1 = Detected  
0 = Not Detected

**CUV (Bit 0):** Cell Undervoltage

1 = Detected  
0 = Not Detected

### 17.2.36 ManufacturerAccess() 0x0052 PFAIert

This command returns the *PFAIert()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                             | Action                                                                                          |
|----------|---------------------------------------|-------------------------------------------------------------------------------------------------|
| Activate | 0x0052 to <i>ManufacturerAccess()</i> | Outputs <i>PFAIert()</i> flags on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

|      |     |     |     |          |          |      |      |       |       |      |     |       |     |     |     |
|------|-----|-----|-----|----------|----------|------|------|-------|-------|------|-----|-------|-----|-----|-----|
| 15   | 14  | 13  | 12  | 11       | 10       | 9    | 8    | 7     | 6     | 5    | 4   | 3     | 2   | 1   | 0   |
| SOTF | TS3 | TS2 | TS1 | AFE_XRDY | AFE_OVRD | AFEC | AFER | DFETF | CFETF | VIMR | SOT | SOCOD | SOC | SOV | SUV |

**SOTF (Bit 15):** Safety Overtemperature FET Failure

1 = Detected  
0 = Not Detected

**TS3 (Bit 14):** Open Thermistor – TS3 Failure

1 = Detected  
0 = Not Detected

**TS2 (Bit 13):** Open Thermistor – TS2 Failure

1 = Detected  
0 = Not Detected

**TS1 (Bit 12):** Open Thermistor – TS1 Failure

1 = Detected

0 = Not Detected

**AFE\_XRDY (Bit 11):** Companion BQ769x0 AFE XREADY Failure

1 = Detected

0 = Not Detected

**AFE\_OVRD (Bit 10):** Companion BQ769x0 AFE Override Failure

1 = Detected

0 = Not Detected

**AFEC (Bit 9):** AFE Communication Failure

1 = Detected

0 = Not Detected

**AFER (Bit 8):** AFE Register Failure

1 = Detected

0 = Not Detected

**DFETF (Bit 7):** Discharge FET Failure

1 = Detected

0 = Not Detected

**CFETF (Bit 6):** Charge FET Failure

1 = Detected

0 = Not Detected

**VIMR (Bit 5):** Voltage Imbalance While Pack Is at Rest Failure

1 = Detected

0 = Not Detected

**SOT (Bit 4):** Safety Overtemperature Cell Failure

1 = Detected

0 = Not Detected

**SOCD (Bit 3):** Safety Overcurrent in Discharge

1 = Detected

0 = Not Detected

**SOCC (Bit 2):** Safety Overcurrent in Charge

1 = Detected

0 = Not Detected

**SOV (Bit 1):** Safety Cell Overvoltage Failure

1 = Detected

0 = Not Detected

**SUV (Bit 0):** Safety Cell Undervoltage Failure

1 = Detected

0 = Not Detected

### 17.2.37 ManufacturerAccess() 0x0053 PFStatus

This command returns the *PFStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                      | Action                                                                                           |
|----------|--------------------------------|--------------------------------------------------------------------------------------------------|
| Activate | 0x0053 to ManufacturerAccess() | Outputs <i>PFStatus()</i> flags on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

|      |      |      |      |          |          |      |      |       |       |      |      |       |      |     |     |
|------|------|------|------|----------|----------|------|------|-------|-------|------|------|-------|------|-----|-----|
| 31   | 30   | 29   | 28   | 27       | 26       | 25   | 24   | 23    | 22    | 21   | 20   | 19    | 18   | 17  | 16  |
| RSVD | RSVD | RSVD | RSVD | RSVD     | RSVD     | RSVD | RSVD | RSVD  | RSVD  | RSVD | RSVD | RSVD  | RSVD | DFW | IFC |
| 15   | 14   | 13   | 12   | 11       | 10       | 9    | 8    | 7     | 6     | 5    | 4    | 3     | 2    | 1   | 0   |
| SOTF | TS3  | TS2  | TS1  | AFE_XRDY | AFE_OVRD | AFEC | AFER | DFETF | CFETF | VIMR | SOT  | SOCOD | SOC  | SOV | SUV |

**RSVD (Bit 31–18):** Reserved. Do not use.

**DFW (Bit 17):** Data Flash Wearout Failure

1 = Detected

0 = Not Detected

**IFC (Bit 16):** Instruction Flash Checksum Failure

1 = Detected

0 = Not Detected

**SOTF (Bit 15):** Safety Overtemperature FET Failure

1 = Detected

0 = Not Detected

**TS3 (Bit 14):** Open Thermistor – TS3 Failure

1 = Detected

0 = Not Detected

**TS2 (Bit 13):** Open Thermistor – TS2 Failure

1 = Detected

0 = Not Detected

**TS1 (Bit 12):** Open Thermistor – TS1 Failure

1 = Detected

0 = Not Detected

**AFE\_XRDY (Bit 11):** Companion BQ769x0 AFE XREADY Failure

1 = Detected

0 = Not Detected

**AFE\_OVRD (Bit 10):** Companion BQ769x0 AFE Override Failure

1 = Detected

0 = Not Detected

**AFEC (Bit 9):** AFE Communication Failure

1 = Detected

0 = Not Detected

**AFER (Bit 8):** AFE Register Failure

1 = Detected

0 = Not Detected

**DFETF (Bit 7):** Discharge FET Failure

1 = Detected

0 = Not Detected

**CFETF (Bit 6):** Charge FET Failure

1 = Detected

0 = Not Detected

**VIMR (Bit 5):** Voltage Imbalance while pack is at rest failure

1 = Detected

0 = Not Detected

**SOT (Bit 4):** Safety Overtemperature Cell Failure

1 = Detected

0 = Not Detected

**SOCD (Bit 3):** Safety Overcurrent in Discharge

1 = Detected

0 = Not Detected

**SOCC (Bit 2):** Safety Overcurrent in Charge

1 = Detected

0 = Not Detected

**SOV (Bit 1):** Safety Cell Overvoltage Failure

1 = Detected

0 = Not Detected

**SUV (Bit 0):** Safety Cell Undervoltage Failure

1 = Detected

0 = Not Detected

### 17.2.38 ManufacturerAccess() 0x0054 OperationStatus

This command returns the *OperationStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| SBS Cmd | Mode | Name            | Format | Size in Bytes | Min    | Max    | Default | Unit |
|---------|------|-----------------|--------|---------------|--------|--------|---------|------|
| 0x0054  | R    | OperationStatus | Hex    | 4             | 0x0000 | 0x00FF | —       | —    |

|       |      |      |    |       |       |      |      |        |      |               |       |      |      |     |      |
|-------|------|------|----|-------|-------|------|------|--------|------|---------------|-------|------|------|-----|------|
| 31    | 30   | 29   | 28 | 27    | 26    | 25   | 24   | 23     | 22   | 21            | 20    | 19   | 18   | 17  | 16   |
| KEYIN | RSVD | RSVD | CB | SLPCC | SLPAD | RSVD | INIT | SLEEPM | XL   | CAL<br>OFFSET | CAL   | RSVD | AUTH | LED | SDM  |
| 15    | 14   | 13   | 12 | 11    | 10    | 9    | 8    | 7      | 6    | 5             | 4     | 3    | 2    | 1   | 0    |
| SLEEP | XCHG | XDSG | PF | SS    | SDV   | SEC1 | SEC0 | RSVD   | RSVD | SAFE          | HCFET | PCHG | DSG  | CHG | PRES |

**KEYIN (Bit 31):** Indicates the state of the KEYIN input detection and is independent of the [KEY\_POL] setting.

1 = KEYIN has been detected.

0 = KEYIN not been detected.

**RSVD (Bits 30–29):** Reserved. Do not use.

**CB (Bit 28):** Cell balancing status

1 = Active

0 = Inactive

**SLPCC (Bit 27):** CC Measurement in SLEEP mode

1 = Active

0 = Inactive

**SLPAD (Bit 26):** ADC Measurement in SLEEP mode

- 1 = Active
- 0 = Inactive

**RSVD (Bit 25):** Reserved. Do not use.

**INIT (Bit 24):** Initialization after full reset

- 1 = Active
- 0 = Inactive

**SLEEPM (Bit 23):** SLEEP mode

- 1 = Active
- 0 = Inactive

**XL (Bit 22):** 400-kHz SMBus mode

- 1 = Active
- 0 = Inactive

**CAL\_OFFSET (Bit 21):** Calibration Output (raw CC offset data).

- 1 = Active when MAC *OutputShortedCCADCCal()* is sent and the raw shorted CC data for calibration is available.
- 0 = When the raw shorted CC data for calibration is not available.

**CAL (Bit 20):** Calibration Output (raw ADC and CC)

- 1 = Active
- 0 = Inactive

**RSVD (Bit 19):** Reserved. Do not use.

**AUTH (Bit 18):** Authentication in progress

- 1 = Active
- 0 = Inactive

**LED (Bit 17):** LED Display

- 1 = LED display is on.
- 0 = LED display is off.

**SDM (Bit 16):** SHUTDOWN triggered via command

- 1 = Active
- 0 = Inactive

**SLEEP (Bit 15):** SLEEP mode conditions met

- 1 = Active
- 0 = Inactive

**XCHG (Bit 14):** Charging disabled

- 1 = Active
- 0 = Inactive

**XDSG (Bit 13):** Discharging disabled

- 1 = Active
- 0 = Inactive

**PF (Bit 12):** PERMANENT FAILURE mode status

- 1 = Active
- 0 = Inactive

**SS (Bit 11):** SAFETY mode status

- 1 = Active

0 = Inactive

**SDV (Bit 10):** Shutdown triggered via low pack voltage

1 = Active

0 = Inactive

**SEC1, SEC0 (Bit 9,8):** SECURITY mode

0, 0 = Reserved

0, 1 = Full Access

1, 0 = Unsealed

1, 1 = Sealed

**RSVD (Bit 6):** Reserved. Do not use.

**SAFE (Bit 5):** SAFE pin status

1 = Active

0 = Inactive

**HCFET (Bit 4):** FETs under control of *HostFETControl()*

1 = FETs under control *HostFETControl()*

0 = FETs are under normal control.

**PCHG (Bit 3):** Precharge FET status

1 = Active

0 = Inactive

**DSG (Bit 2):** DSG FET status

1 = Active

0 = Inactive

**CHG (Bit 1):** CHG FET status

1 = Active

0 = Inactive

**PRES (Bit 0):** System present low

1 = Active

0 = Inactive

### 17.2.39 ManufacturerAccess() 0x0055 ChargingStatus

This command returns the *ChargingStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                             | Action                                                                                                                                          |
|----------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0055 to <i>ManufacturerAccess()</i> | Outputs <i>ChargingStatus()</i> flags (MSB) AND <i>TempRange()</i> flags (LSB) on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

|     |      |    |    |      |      |      |      |      |      |      |    |    |    |    |    |
|-----|------|----|----|------|------|------|------|------|------|------|----|----|----|----|----|
| 15  | 14   | 13 | 12 | 11   | 10   | 9    | 8    | 7    | 6    | 5    | 4  | 3  | 2  | 1  | 0  |
| VCT | RSVD | SU | IN | RSVD | RSVD | FCHG | PCHG | RSVD | RSVD | RSVD | OT | HT | ST | LT | UT |

**VCT (Bit 15):** Charge Termination

1 = Active

0 = Inactive

**RSVD (Bit 14):** Reserved. Do not use.

**SU (Bit 13): Charge Suspend**

1 = Active  
0 = Inactive

**IN (Bit 12): Charge Inhibit**

1 = Active  
0 = Inactive

**RSVD (Bits 11–10): Reserved. Do not use.**
**FCHG (Bit 9): FAST-CHARGE mode**

1 = Active  
0 = Inactive

**PCHG (Bit 8): PRE-CHARGE mode**

1 = Active  
0 = Inactive

**RSVD (Bits 7–5): Reserved. Do not use.**
**OT (Bit 4): Overtemperature Region**

1 = Active  
0 = Inactive

**HT (Bit 3): High Temperature Region**

1 = Active  
0 = Inactive

**ST (Bit 2): Standard Temperature Region**

1 = Active  
0 = Inactive

**LT (Bit 1): Low Temperature Region**

1 = Active  
0 = Inactive

**UT (Bit 0): Undertemperature Region**

1 = Active  
0 = Inactive

### 17.2.40 ManufacturerAccess() 0x0056 GaugingStatus

This command returns the *GaugingStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                             | Action                                                                                                  |
|----------|---------------------------------------|---------------------------------------------------------------------------------------------------------|
| Activate | 0x0056 to <i>ManufacturerAccess()</i> | Outputs <i>GaugingStatus()</i> flags on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> . |

|     |      |      |      |      |      |       |      |    |     |      |        |    |    |    |    |
|-----|------|------|------|------|------|-------|------|----|-----|------|--------|----|----|----|----|
| 15  | 14   | 13   | 12   | 11   | 10   | 9     | 8    | 7  | 6   | 5    | 4      | 3  | 2  | 1  | 0  |
| VDQ | EDV2 | EDV1 | RSVD | RSVD | FCCX | OCVFR | REST | CF | DSG | EDV0 | BAL_OK | TC | TD | FC | FD |

**VDQ (Bit 15): Discharge Qualified for Learning (based on the RU flag)**

1 = Detected  
0 = Not Detected

**EDV2 (Bit 14): End-of-Discharge Voltage Level 2**



- 1 = EDV2 voltage is reached during discharge.
- 0 = EDV2 voltage is not reached, or not in DISCHARGE mode.

**EDV1 (Bit 13):** End-of-Discharge Voltage Level 1

- 1 = EDV1 voltage is reached during discharge.
- 0 = EDV1 voltage is not reached, or not in DISCHARGE mode.

**RSVD (Bits 12–11):** Reserved. Do not use.

**FCCX (Bit 10):** *FullChargeCapacity()* has been updated. This bit changes state each time *FullChargeCapacity()* updates.

**OCVFR (Bit 9):** Reserved and not used.

**REST (Bit 8):** Rest

- 1 = OCV Reading Taken
- 0 = OCV Reading Not Taken or Not in Relax

**CF (Bit 7):** Condition Flag

- 1 = *MaxError()* > Max Error Limit (Condition Cycle needed)
- 0 = *MaxError()* < Max Error Limit (Condition Cycle not needed)

**DSG (Bit 6):** Discharge/Relax

- 1 = Charging Not Detected
- 0 = Charging Detected

**EDV0 (Bit 5):** End-of-Discharge Voltage Level 0 (Termination)

- 1 = Termination voltage is reached during discharge.
- 0 = Termination voltage is not reached, or not in DISCHARGE mode.

**BAL\_OK (Bit 4):** Cell Balancing

- 1 = Cell balancing is possible if enabled.
- 0 = Cell balancing is not allowed.

**TC (Bit 3):** Terminate Charge

- 1 = Detected
- 0 = Not Detected

**TD (Bit 2):** Terminate Discharge

- 1 = Detected
- 0 = Not Detected

**FC (Bits 1):** Fully Charged

- 1 = Detected
- 0 = Not Detected

**FD (Bit 0):** Fully Discharged

- 1 = Detected
- 0 = Not Detected

### 17.2.41 ManufacturerAccess() 0x0057 ManufacturingStatus

This command returns the *ManufacturingStatus()* flags on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                             | Action                                                                                                        |
|----------|---------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Activate | 0x0057 to <i>ManufacturerAccess()</i> | Outputs <i>ManufacturingStatus()</i> flags on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> . |

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|        |         |         |             |      |          |          |           |
|--------|---------|---------|-------------|------|----------|----------|-----------|
| 15     | 14      | 13      | 12          | 11   | 10       | 9        | 8         |
| CAL_EN | LT_TEST | CB_TEST | AFE_DD_TEST | RSVD | RSVD     | LED_EN   | SAFE_EN   |
| 7      | 6       | 5       | 4           | 3    | 2        | 1        | 0         |
| BBR_EN | PF_EN   | LF_EN   | FET_EN      | RSVD | DSG_TEST | CHG_TEST | PCHG_TEST |

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

**CAL\_EN (Bit 15):** CALIBRATION mode

- 1 = Enabled
- 0 = Disabled

**LT\_TEST (Bit 14):** LIFETIME SPEED UP mode

- 1 = Enabled
- 0 = Disabled

**CB\_TEST (Bit 13):** Cell Balancing Test

- 1 = Enabled
- 0 = Disabled

**AFE\_DD\_TEST (Bit 12):** AFE Delay Disable Test

- 1 = Enabled
- 0 = Disabled

**RSVD (Bits 11–10):** Reserved. Do not use.

**LED\_EN (Bit 9):** LED Display

- 1 = Enabled
- 0 = Disabled

**SAFE\_EN (Bit 8):** SAFE Action

- 1 = Enabled
- 0 = Disabled

**BBR\_EN (Bit 7):** Black Box Recorder

- 1 = Enabled
- 0 = Disabled

**PF\_EN (Bit 6):** Permanent Failure

- 1 = Enabled
- 0 = Disabled

**LF\_EN (Bit 5):** Lifetime Data Collection

- 1 = Enabled
- 0 = Disabled

**FET\_EN (Bit 4):** All FET Action

- 1 = Enabled
- 0 = Disabled

**RSVD (Bit 3):** Reserved. Do not use.

**DSG\_TEST (Bit 2):** Discharge FET Test

- 1 = Discharge FET test activated
- 0 = Disabled

**CHG\_TEST (Bit 1):** Charge FET Test

- 1 = Charge FET test activated
- 0 = Disabled

**PCHG\_TEST (Bit 0): Precharge FET Test**

1 = Precharge FET test activated

0 = Disabled

**17.2.42 ManufacturerAccess() 0x0058 AFESatus**

This command returns the configuration of the companion AFE (address 0x00). See the AFE *BQ769x0 3-Series to 15-Series Cell Battery Monitor Family for Li-Ion and Phosphate Applications Data Manual* ([SLUSBK2](#)) for further details.

| SBS Cmd | Mode | Name     | Format | Size in Bytes | Min    | Max    | Default | Unit |
|---------|------|----------|--------|---------------|--------|--------|---------|------|
| 0x0058  | R    | AFESatus | Hex    | 2             | 0x0000 | 0x00FF | —       | —    |

**17.2.43 ManufacturerAccess() 0x0059 AFEConfig**

This command returns the configuration of the companion AFE (address 0x01 to 0x0B). See the AFE *BQ769x0 3-Series to 15-Series Cell Battery Monitor Family for Li-Ion and Phosphate Applications Data Manual* ([SLUSBK2](#)).

| SBS Cmd | Mode | Name      | Format | Size in Bytes | Min | Max | Default | Unit |
|---------|------|-----------|--------|---------------|-----|-----|---------|------|
| 0x0059  | R    | AFEConfig | String | 11+1          | —   | —   | —       | —    |

**17.2.44 ManufacturerAccess() 0x005A AFEVCx**

This command returns the cell voltage measurement data of the companion AFE (address 0x0C to 0x29) on *ManufacturerBlockAccess()* or *ManufacturerData()*. See the AFE *BQ769x0 3-Series to 15-Series Cell Battery Monitor Family for Li-Ion and Phosphate Applications Data Manual* ([SLUSBK2](#)).

| SBS Cmd | Mode | Name   | Format | Size in Bytes | Min | Max | Default | Unit |
|---------|------|--------|--------|---------------|-----|-----|---------|------|
| 0x005A  | R    | AFEVCx | String | 30+1          | —   | —   | —       | —    |

**17.2.45 ManufacturerAccess() 0x005B AFEData**

This command returns the system voltage, temperature, and current measurement data of the companion AFE (address 0x2A to 0x33, 0x50, 0x51, and 0x59) on *ManufacturerBlockAccess()* or *ManufacturerData()*. See the AFE *BQ769x0 3-Series to 15-Series Cell Battery Monitor Family for Li-Ion and Phosphate Applications Data Manual* ([SLUSBK2](#)).

| SBS Cmd | Mode | Name    | Format | Size in Bytes | Min | Max | Default | Unit |
|---------|------|---------|--------|---------------|-----|-----|---------|------|
| 0x005B  | R    | AFEData | String | 13+1          | —   | —   | —       | —    |

**17.2.46 ManufacturerAccess() 0x0060 Lifetime Data Block 1**

This command returns the Lifetime data on *ManufacturerBlockAccess()* or *ManufacturerData()*. See for details.

| Status   | Condition                             | Action                                                                                                            |
|----------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0060 to <i>ManufacturerAccess()</i> | Outputs lifetime data values of Max Cell Voltage on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

### 17.2.47 ManufacturerAccess() 0x0061 Lifetime Data Block 2

This command returns the Lifetime data on *ManufacturerBlockAccess()* or *ManufacturerData()*. See [Lifetimes](#) for details.

| Status   | Condition                             | Action                                                                                                            |
|----------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0061 to <i>ManufacturerAccess()</i> | Outputs lifetime data values of Min Cell Voltage on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

### 17.2.48 ManufacturerAccess() 0x0062 Lifetime Data Block 3

This command returns the Lifetime data on *ManufacturerBlockAccess()* or *ManufacturerData()*. See [Lifetimes](#) for details.

| Status   | Condition                             | Action                                                                                                                                           |
|----------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0062 to <i>ManufacturerAccess()</i> | Outputs lifetime data values of Delta Voltage, Currents, Power and Temperatures on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

### 17.2.49 ManufacturerAccess() 0x0063 Lifetime Data Block 4

This command returns the Lifetime data on *ManufacturerBlockAccess()* or *ManufacturerData()*. See [Lifetimes](#) for details.

| Status   | Condition                             | Action                                                                                                                                                          |
|----------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0063 to <i>ManufacturerAccess()</i> | Outputs lifetime data of the Power events and Cell Balancing Times data on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> . (OLD, SCD, OTC, OTD) |

### 17.2.50 ManufacturerAccess() 0x0064 Lifetime Data Block 5

This command returns the Lifetime data on *ManufacturerBlockAccess()* or *ManufacturerData()*. See [Lifetimes](#) for details.

| Status   | Condition                             | Action                                                                                                    |
|----------|---------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Activate | 0x0064 to <i>ManufacturerAccess()</i> | Outputs lifetime data of the Time values on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

### 17.2.51 ManufacturerAccess() 0x0065 Lifetime Data Block 6

This command returns the Lifetime data on *ManufacturerBlockAccess()* or *ManufacturerData()*. See [Lifetimes](#) for details.

| Status   | Condition                             | Action                                                                                                                                |
|----------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0065 to <i>ManufacturerAccess()</i> | Outputs lifetime data of the safety events data on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> (COV, CUV, OCD, OCC) |

### 17.2.52 ManufacturerAccess() 0x0066 Lifetime Data Block 7

This command returns the Lifetime data on *ManufacturerBlockAccess()* or *ManufacturerData()*. See [Lifetimes](#) for details.

| Status   | Condition                             | Action                                                                                                                                                             |
|----------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0066 to <i>ManufacturerAccess()</i> | Outputs lifetime data of the charger termination and FC update events data on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> . (OLD, SCD, OTC, OTD) |

### 17.2.53 ManufacturerAccess() 0x0070 ManufacturerInfo

This command returns **ManufacturerInfo** on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                             | Action                                                                                                                                                                                                    |
|----------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0070 to <i>ManufacturerAccess()</i> | Outputs 32 bytes of ManufacturerInfo on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> in the following format:<br>AABBCCDDEEFFGGHHIIJJKKLLMMNNNOOPPQQRRSST<br>TUUUVVWWXXYYZZ112233 445566 |

### 17.2.54 ManufacturerAccess() 0x0071 DStatus1

This command returns the CellVoltages on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                             | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0071 to <i>ManufacturerAccess()</i> | Outputs 32 bytes of data on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> in the following format:<br>aaAAbbBBccCCddDDeeEEffFFggGGhhHHiiIjJJkkKKllLL<br>mmMMnnNNooOOppPP where:<br>AAaa: Cell Voltage 1<br>BBbb: Cell Voltage 2<br>CCcc: Cell Voltage 3<br>DDdd: Cell Voltage 4<br>EEee: Cell Voltage 5<br>FFff: Cell Voltage 6<br>GGgg: Cell Voltage 7<br>HHhh: Cell Voltage 8<br>IIii: Cell Voltage 9<br>JJjj: Cell Voltage 10<br>KKkk: Cell Voltage 11<br>LLll: Cell Voltage 12<br>MMmm: Cell Voltage 13<br>NNnn: Cell Voltage 14<br>OOoo: Cell Voltage 15<br>PPpp: Reserved |

### 17.2.55 ManufacturerAccess() 0x0072 DStatus2

This command returns the TS1, TS2, TS3, and cell temperature data on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| Status   | Condition                             | Action                                                                                                                                                                                                                                                                                                                                                                           |
|----------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0072 to <i>ManufacturerAccess()</i> | Outputs 16 bytes of temperature data values on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> in the following format: aaAAbbBBccCCddDD where:<br>AAaa: ExtAveCellVoltage<br>BBbb: VAUX Voltage<br>CCcc: TS1 Temperature<br>DDdd: TS2 Temperature<br>EEee: TS3 Temperature<br>FFff: Cell temperature<br>GGgg: FET Temperature<br>HHhh: Internal Gauge Temperature |

### 17.2.56 ManufacturerAccess() 0x0080 CUV Snapshot

This command returns the CellVoltages on *ManufacturerBlockAccess()* or *ManufacturerData()* at the time *SafetyStatus()* [CUV] is set. It is refreshed each time [CUV] becomes set.

| Status   | Condition                      | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0080 to ManufacturerAccess() | Outputs 32 bytes of data on ManufacturerBlockAccess() or ManufacturerData() in the following format:<br>aaAAbbBBccCCddDDeeEEffFFggGGhhHHiiIJJjKkKKIILLmmMMnnNNooOOppPP where:<br>AAaa: Cell Voltage 1<br>BBbb: Cell Voltage 2<br>CCcc: Cell Voltage 3<br>DDdd: Cell Voltage 4<br>EEee: Cell Voltage 5<br>FFff: Cell Voltage 6<br>GGgg: Cell Voltage 7<br>HHhh: Cell Voltage 8<br>Iiii: Cell Voltage 9<br>JJjj: Cell Voltage 10<br>KKkk: Cell Voltage 11<br>LLll: Cell Voltage 12<br>MMmm: Cell Voltage 13<br>NNnn: Cell Voltage 14<br>OOoo: Cell Voltage 15<br>PPpp: Reserved |

### 17.2.57 ManufacturerAccess() 0x0081 COV Snapshot

This command returns the CellVoltages on ManufacturerBlockAccess() or ManufacturerData() at the time SafetyStatus() [COV] is set. It is refreshed each time [COV] becomes set.

| Status   | Condition                      | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate | 0x0081 to ManufacturerAccess() | Outputs 32 bytes of data on ManufacturerBlockAccess() or ManufacturerData() in the following format:<br>aaAAbbBBccCCddDDeeEEffFFggGGhhHHiiIJJjKkKKIILLmmMMnnNNooOOppPP where:<br>AAaa: Cell Voltage 1<br>BBbb: Cell Voltage 2<br>CCcc: Cell Voltage 3<br>DDdd: Cell Voltage 4<br>EEee: Cell Voltage 5<br>FFff: Cell Voltage 6<br>GGgg: Cell Voltage 7<br>HHhh: Cell Voltage 8<br>Iiii: Cell Voltage 9<br>JJjj: Cell Voltage 10<br>KKkk: Cell Voltage 11<br>LLll: Cell Voltage 12<br>MMmm: Cell Voltage 13<br>NNnn: Cell Voltage 14<br>OOoo: Cell Voltage 15<br>PPpp: Reserved |

### 17.2.58 ManufacturerAccess() 0x0F00 ROM Mode

This command sends the device into ROM mode in preparation for re-programming.

| Status   | Condition                                                             | Action                                                                               |
|----------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| ROM Mode | OperationStatus()[SEC1,SEC0] = 0,1 AND 0x0F00 to ManufacturerAccess() | Device goes to ROM mode ready for update, use 0x08 to ManufacturerAccess() to return |

### 17.2.59 Data Flash Access() 0x4000–0x5FFF

Accessing data flash is only supported by the ManufacturerBlockAccess() by addressing the physical address.

To write to the DF, send the starting address, followed by the DF data block. The DF data block is the intended revised DF data to be updated to DF. The size of the DF data block ranges from 1 byte to 32 bytes. All individual data must be sent in Little Endian.

Write to DF example:

Assuming: data1 locates at address 0x4000 and data2 locates at address 0x4002.

Both data1 and data2 are U2 type.

To update data1 and data2, send a SMBus block write with command = 0x44

block = starting address + DF data block

= 0x00 + 0x40 + data1\_LowByte + data1\_HighByte + data2\_LowByte + data2\_HighByte

To read the DF, send an SMBus block write to the *ManufacturerBlockAccess()*, followed by the starting address; then send an SMBus block read to the *ManufacturerBlockAccess()*. The return data contains the starting address, followed by 32 bytes of DF data in Little Endian.

Read from DF example:

Taking the same assuming from the read DF example, to read DF,

a. Send SMBus write block with command 0x44, block = 0x00 + 0x40

b. Send SMBus read block with command 0x44

The returned block

= a starting address + 32 bytes of DF dat

= 0x00 + 0x40 + data1\_LowByte + data1\_HighByte + data2\_LowByte + data2\_HighByte.... data32\_LowByte + data32\_HighByte

The gauge supports an auto-increment on the address during a DF read. This greatly reduces the time required to read out the entire DF. Continue with the read from the DF example. If another SMBus read block is sent with command 0x44, the gauge returns another 32 bytes of DF data, starting with address 0x4020.

### 17.2.60 ManufacturerAccess() 0xF080 Exit Calibration Output Mode

This command stops the output of calibration data to the *ManufacturerBlockAccess()* or *ManufacturerData()* command.

| Status   | Condition                                                                                                         | Action                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Activate | <i>ManufacturerBlockAccess()</i> or<br><i>ManufacturerData()</i> = 1 AND 0xF080 to<br><i>ManufacturerAccess()</i> | Stops output of ADC or CC data on<br><i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> |

### 17.2.61 ManufacturerAccess() 0xF081 OutputCellVoltageforCalibration

This command instructs the device to output the raw values for calibration purposes on *ManufacturerBlockAccess()* or *ManufacturerData()*. All values are updated every 250 ms and the format of each value is 2's complement, MSB first.

| Status  | Condition                                                                        | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disable | <i>ManufacturingStatus()</i> [CAL] = 1 AND 0xF080 to <i>ManufacturerAccess()</i> | <i>ManufacturingStatus()</i> [CAL] = 0<br>Stops output of ADC and CC data on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Enable  | 0xF081 to <i>ManufacturerAccess()</i>                                            | <i>ManufacturingStatus()</i> [CAL] = 1<br>Outputs the raw CC and AD values<br><i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> in the format of ZZYyaaAAbbBBccCCddDDeeEEffFF<br>ggGGhhHHiiIjjJkkKK:<br>ZZ: rolling 8-bit counter, increments when values are refreshed.<br>YY: Status: 1 when <i>ManufacturerAccess()</i> = 0xF081;<br>2 when <i>ManufacturerAccess()</i> = 0xF082<br>AAaa: AFE Cell Map<br>BBbb: Cell Voltage 1<br>CCcc: Cell Voltage 2<br>DDdd: Cell Voltage 3<br>EEee: Cell Voltage 4<br>FFff: Cell Voltage 5<br>GGgg: Cell Voltage 6<br>HHhh: Cell Voltage 7<br>IIii: Cell Voltage 8<br>JJjj: Cell Voltage 9<br>KKkk: Cell Voltage 10<br>LLll: Cell Voltage 11<br>MMmm: Cell Voltage 12<br>NNnn: Cell Voltage 13<br>OOoo: Cell Voltage 14 |

### 17.2.62 *ManufacturerAccess()* 0xF082 *OutputCellVoltageCCandTempforCalibration*

This command instructs the device to output the raw values for calibration purposes on *ManufacturerBlockAccess()* or *ManufacturerData()*. All values are updated every 250 ms and the format of each value is 2's complement, MSB first. This mode includes an internal short on the coulomb counter inputs for measuring offset.

| Status  | Condition                                                                        | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disable | <i>ManufacturingStatus()</i> [CAL] = 1 AND 0xF080 to <i>ManufacturerAccess()</i> | <i>ManufacturingStatus()</i> [CAL] = 0<br>Stop output of ADC and CC data on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Enable  | 0xF081 to <i>ManufacturerAccess()</i>                                            | <i>ManufacturingStatus()</i> [CAL] = 1<br>Outputs the raw CC and AD values on <i>ManufacturerBlockAccess()</i> or <i>ManufacturerData()</i> in the format of ZZYyaaAAbbBBccCCddDDeeEEffFF<br>ggGGhhHHiiIjjJkkKK:<br>ZZ: rolling 8-bit counter, increments when values are refreshed.<br>YY: Status: 1 when <i>ManufacturerAccess()</i> = 0xF081;<br>2 when <i>ManufacturerAccess()</i> = 0xF082<br>AAaa: AFE Cell Map<br>BBbb: Cell Voltage 15<br>CCcc: Reserved<br>DDdd: ExtAveCellVoltage<br>EEeeFFff: VAUX Voltage<br>GGgg: Current (Coulomb Counter)<br>HHhh: TS1 Temperature<br>IIii: TS2 Temperature<br>JJjj: TS3 Temperature<br>KKkk: Gauge Temperature |



### 17.3 0x01 RemainingCapacityAlarm()

This read- or write-word function sets or gets a low-capacity alarm threshold value. The default value for *RemainingCapacityAlarm* is stored in **Rem Cap Alarm**. If *RemainingCapacityAlarm()* is set to 0, the alarm is disabled. If *RemainingCapacity()* < *RemainingCapacityAlarm()*, the [RCA] flag is set and the BQ78350-R1A sends an *AlarmWarning()* message to the SMBUS host. If *RemainingCapacity()* ≥ *RemainingCapacityAlarm()* and [DSG] is set, the [RCA] flag is cleared. 0 = remaining capacity alarm is disabled. 1..700 = the remaining capacity limit for [RCA] flag.

| SBS Cmd | Name                            | Access |    |    | Protocol | Type | Min | Max | Default | Unit   | Note                            |
|---------|---------------------------------|--------|----|----|----------|------|-----|-----|---------|--------|---------------------------------|
|         |                                 | SE     | US | FA |          |      |     |     |         |        |                                 |
| 0x01    | <i>RemainingCapacityAlarm()</i> | R/W    |    |    | Word     | U2   | 0   | 700 | 300     | mAh    | <i>BatteryMode()</i> [CAPM] = 0 |
|         |                                 |        |    |    |          |      |     |     |         | 10 mWh | <i>BatteryMode()</i> [CAPM] = 1 |

The threshold value is stored in the following data flash location.

| Class             | Subclass | Name                     | Type | Min | Max  | Default | Unit | Description                                     |
|-------------------|----------|--------------------------|------|-----|------|---------|------|-------------------------------------------------|
| SBS Configuration | Data     | Remaining Capacity Alarm | U2   | 0   | 700  | 300     | mAh  | <i>RemainingCapacityAlarm()</i> value in mAh    |
| SBS Configuration | Data     | Remaining Capacity Alarm | U2   | 0   | 1000 | 432     | cWh  | <i>RemainingCapacityAlarm()</i> value in 10 mWh |

### 17.4 0x02 RemainingTimeAlarm()

This read- or write-word function sets or gets the *RemainingTimeAlarm()* value. The default value of *RemainingTimeAlarm()* is stored in **Rem Time Alarm**. If *RemainingTimeAlarm()* = 0, this alarm is disabled. If *AverageTimeToEmpty()* < *RemainingTimeAlarm()*, the [RTA] flag is set and the BQ78350-R1A sends an *AlarmWarning()* message to the SMBus host. If *AverageTimeToEmpty()* ≥ *RemainingTimeAlarm()*, the [RTA] flag is reset.

0 = remaining time alarm is disabled. 1..30 = the remaining time limit for the [RTA] flag.

| SBS Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit |
|---------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|
|         |                             | SE     | US | FA |          |      |     |     |         |      |
| 0x02    | <i>RemainingTimeAlarm()</i> | R/W    |    |    | Word     | U2   | 0   | 30  | 10      | min  |

The threshold value is stored in the following data flash location.

| Class             | Subclass | Name                 | Type | Min | Max | Default | Unit | Description                       |
|-------------------|----------|----------------------|------|-----|-----|---------|------|-----------------------------------|
| SBS Configuration | Data     | Remaining Time Alarm | U2   | 0   | 30  | 10      | min  | <i>RemainingTimeAlarm()</i> value |

### 17.5 0x03 BatteryMode()

This read- or write-word function selects the various battery operational modes, reports the battery's capabilities and modes, and flags minor conditions that require attention.

| SBS Cmd | Name                 | Access |    |    | Protocol | Type | Min    | Max    | Unit |
|---------|----------------------|--------|----|----|----------|------|--------|--------|------|
|         |                      | SE     | US | FA |          |      |        |        |      |
| 0x03    | <i>BatteryMode()</i> | R/W    |    |    | Word     | H2   | 0x0000 | 0xFFFF | —    |

|      |      |    |      |      |      |    |    |    |      |      |      |      |      |     |     |
|------|------|----|------|------|------|----|----|----|------|------|------|------|------|-----|-----|
| 15   | 14   | 13 | 12   | 11   | 10   | 9  | 8  | 7  | 6    | 5    | 4    | 3    | 2    | 1   | 0   |
| CAPM | CHGM | AM | RSVD | RSVD | RSVD | PB | CC | CF | RSVD | RSVD | RSVD | RSVD | RSVD | PBS | ICC |

**CAPM (Bit 15):** CAPACITY Mode (R/W)

- 1 = Reports specific data in 10 mW or 10 mWh
- 0 = Reports specific data in mA or mAh (default)

**CHGM (Bit 14):** CHARGER Mode (R/W)

- 1 = Disable *ChargingVoltage()* and *ChargingCurrent()* broadcasts to host and smart battery charger (default)
- 0 = Enable *ChargingVoltage()* and *ChargingCurrent()* broadcasts to host and smart battery charger

**AM (Bit 13):** ALARM Mode (R/W)

- 1 = Disable Alarm Warning broadcasts to host and smart battery charger (default)
- 0 = Enable AlarmWarning broadcasts to host and smart battery charger

**PB (Bit 9):** Primary Battery. This bit does not affect the operation of the BQ78350-R1A device and is for information only.

- 1 = Battery is operating in its secondary role.
- 0 = Battery is operating in its primary role (default).

**CC (Bit 8):** Charge Controller Enabled (R/W). This bit does not affect the operation of the BQ78350-R1A device and is for information only.

- 1 = Internal charge control enabled
- 0 = Internal charge controller disabled (default)

**CF (Bit 7):** Condition Flag (R). This bit a the same as *GaugingStatus()* [CF]

- 1 = Conditioning cycle requested
- 0 = Battery OK

**PBS (Bit 1):** Primary Battery Support (R). This bit does not affect the operation of the BQ78350-R1A device and is for information only.

- 1 = Primary or Secondary Battery Support
- 0 = Function not supported (default)

**ICC (Bit 0):** Internal Charge Controller (R). This bit does not affect the operation of the BQ78350-R1A device and is for information only.

- 1 = Function supported (default)
- 0 = Function not supported

| Class             | Subclass | Name                 | Format | Size in Bytes | Min    | Max    | Default | Unit |
|-------------------|----------|----------------------|--------|---------------|--------|--------|---------|------|
| SBS Configuration | Data     | Initial Battery Mode | Hex    | 2             | 0x0000 | 0xFFFF | 0x81    | —    |

## 17.6 0x04 *AtRate()*

This read- or write-word function is the first half of a two-function call set used to set the *AtRate* value, which is used in calculations made by the *AtRateTimeToFull()*, *AtRateTimeToEmpty()*, and *AtRateOK()* functions.

The *AtRate()* units are in either mA (*[CapM]* = 0) or 10 mW (*[CapM]* = 1). When the *AtRate()* value is positive, the *AtRateTimeToFull()* function returns the predicted time to full charge at the *AtRate()* value of charge. When the *AtRate()* value is negative, the *AtRateTimeToEmpty()* function returns the predicted operating time at the *AtRate()* value of discharge. When the *AtRate()* value is negative, the *AtRateOK()* function returns a Boolean value that predicts the battery's ability to supply the *AtRate()* value of additional discharge energy (current or power) for 10 s.

| SBS<br>Cmd | Name            | Access |    |    | Proto-<br>col | Type | Min    | Max   | Default | Unit   | Note                            |
|------------|-----------------|--------|----|----|---------------|------|--------|-------|---------|--------|---------------------------------|
|            |                 | SE     | US | FA |               |      |        |       |         |        |                                 |
| 0x04       | <i>AtRate()</i> | R/W    |    |    | Word          | I2   | –32768 | 32767 | 0       | mAh    | <i>BatteryMode()</i> [CAPM] = 0 |
|            |                 |        |    |    |               |      |        |       |         | 10 mWh | <i>BatteryMode()</i> [CAPM] = 1 |

## 17.7 0x05 *AtRateTimeToFull()*

This read-word function returns an unsigned integer value of the predicted remaining time to fully charge the battery using a CC-CV method at the *AtRate()* value in minutes with a range of 0 to 65,534. A value of 65,535 indicates that the *AtRate()* = 0.

*AtRateTimeToFull()* can report time based on constant current ([*CapM*] = 0) or constant power ([*CapM*] = 1), and is updated within 1 s after the SMBus host sets the *AtRate()* value. The BQ78350-R1A automatically updates *AtRateTimeToFull()* based on the *AtRate()* function at 1-s intervals. 0..65,534 = predicted time to full charge, based on *AtRate()*. 65,535 = no charge or discharge (*AtRate()* is 0).

| SBS<br>Cmd | Name                      | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit | Note                               |
|------------|---------------------------|--------|----|----|---------------|------|-----|-------|------|------------------------------------|
|            |                           | SE     | US | FA |               |      |     |       |      |                                    |
| 0x05       | <i>AtRateTimeToFull()</i> | R      |    |    | Word          | U2   | 0   | 65535 | min  | 65535 indicates not being charged. |

## 17.8 0x06 *AtRateTimeToEmpty()*

This read-word function returns an unsigned integer value of the predicted remaining operating time in minutes with a range of 0 to 65,534 if the battery is discharged at the *AtRate()* value. A value of 65,535 indicates that *AtRate()* = 0.

*AtRateTimeToEmpty()* can report time based on constant current ([*LDMD*] = 0) or constant power ([*LDMD*] = 1), and is updated within 1 s after the SMBus host sets the *AtRate()* value. The BQ78350-R1A updates *AtRateTimeToEmpty()* at 1-s intervals.

0..65,534 = predicted remaining operating time, based on *AtRate()*. 65,535 = no charge or discharge (*AtRate()* is 0).

| SBS<br>Cmd | Name                       | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit | Note                               |
|------------|----------------------------|--------|----|----|---------------|------|-----|-------|------|------------------------------------|
|            |                            | SE     | US | FA |               |      |     |       |      |                                    |
| 0x06       | <i>AtRateTimeToEmpty()</i> | R      |    |    | Word          | U2   | 0   | 65535 | min  | 65535 indicates not being charged. |

## 17.9 0x07 *AtRateOK()*

This read-word function returns a boolean value that indicates whether or not the battery can deliver the *AtRate()* value of energy for 10 seconds. The BQ78350-R1A updates this value within 1 s after the SMBus host sets the *AtRate()* function value. The BQ78350-R1A updates *AtRateOK()* at 1-s intervals.

If the *AtRate()* function returns  $\geq 0$ , *AtRateOK()* always returns TRUE. Based on the discharge rate indicated in *AtRate()*, if it returns 0, 0 = FALSE and BQ78350-R1A **cannot** deliver energy for 10 seconds. 1..65,535 = TRUE and BQ78350-R1A **can** deliver energy for 10 seconds.

| SBS<br>Cmd | Name              | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit | Note              |
|------------|-------------------|--------|----|----|---------------|------|-----|-------|------|-------------------|
|            |                   | SE     | US | FA |               |      |     |       |      |                   |
| 0x07       | <i>AtRateOK()</i> | R      |    |    | Word          | U2   | 0   | 65535 | —    | 0 = No<br>1 = Yes |

## 17.10 0x08 *Temperature()*

This read-word function returns the temperature in units of 0.1 K, as measured by the BQ769x0 AFE. The source of the measured temperature is configured in **Settings: Temperature Enable**. The selection of Max or Average, if multiple temp sensors are enabled, is configured by **DA Configuration[CTEMP]**.

| SBS Cmd | Name                 | Access |    |    | Protocol | Type | Min | Max   | Unit  |
|---------|----------------------|--------|----|----|----------|------|-----|-------|-------|
|         |                      | SE     | US | FA |          |      |     |       |       |
| 0x08    | <i>Temperature()</i> | R      |    |    | Word     | U2   | 0   | 65535 | 0.1 K |

### 17.11 0x09 Voltage()

This read-word function returns the sum of the measured cell voltages and is scaled per *SpecificationInfo()*.

| SBS Cmd | Name             | Access |    |    | Protocol | Type | Min | Max   | Unit |
|---------|------------------|--------|----|----|----------|------|-----|-------|------|
|         |                  | SE     | US | FA |          |      |     |       |      |
| 0x09    | <i>Voltage()</i> | R      |    |    | Word     | U2   | 0   | 65535 | mV   |

### 17.12 0x0A Current()

This read-word function returns an integer value of the measured current being supplied (or accepted) by the battery in mA, with a range of –32,768 to 32,767. A positive value indicates charge current and a negative value indicates discharge. Any current value within **Deadband** is reported as 0 mA by the *Current()* function.

| SBS Cmd | Name             | Access |    |    | Protocol | Type | Min    | Max   | Unit |
|---------|------------------|--------|----|----|----------|------|--------|-------|------|
|         |                  | SE     | US | FA |          |      |        |       |      |
| 0x0A    | <i>Current()</i> | R      |    |    | Word     | I2   | –32767 | 32768 | mA   |

The value reported is an average of four readings of the CC\_HI and CC\_LO registers of the companion AFE taken at 250-ms intervals.

### 17.13 0x0B AverageCurrent()

This read-word function returns a signed integer value that approximates a one-minute rolling average of the current being supplied (or accepted) through the battery terminals in mA, with a range of –32,768 to 32,767.

*AverageCurrent()* is calculated by a rolling IIR filtered average of *Current()* function data with a period of 14.5 s. During the time after a reset or when switching between charge and discharge currents and before 14.5 s has elapsed, the reported *AverageCurrent()* = *Current()* function value.

| SBS Cmd | Name                    | Access |    |    | Protocol | Type | Min    | Max   | Unit |
|---------|-------------------------|--------|----|----|----------|------|--------|-------|------|
|         |                         | SE     | US | FA |          |      |        |       |      |
| 0x0B    | <i>AverageCurrent()</i> | R      |    |    | Word     | I2   | –32767 | 32768 | mA   |

### 17.14 0x0C MaxError()

This read-word function returns the expected margin of error, in %, in the state-of-charge calculation with a range of 1 to 100%.

| SBS Cmd | Name              | Access |    |    | Protocol | Type | Min | Max  |
|---------|-------------------|--------|----|----|----------|------|-----|------|
|         |                   | SE     | US | FA |          |      |     |      |
| 0x0C    | <i>MaxError()</i> | R      |    |    | Word     | U1   | 0%  | 100% |

| Status Condition                                         | Action                   |
|----------------------------------------------------------|--------------------------|
| Full device reset                                        | <i>MaxError()</i> = 100% |
| FCC updated                                              | <i>MaxError()</i> = 2%   |
| FCC Updated but capped by FCC Learn Up or FCC Learn Down | <i>MaxError()</i> = 8%   |

| Status Condition                                                                                        | Action                                                    |
|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Each <i>CycleCount()</i> increment after last valid FCC update                                          | <i>MaxError()</i> increment by 0.25% to a maximum of 100% |
| After <b>Requested Learning Cycle Count</b> increments of <i>CycleCount()</i> since the last FCC update | <i>CEDV GaugingStatus()</i> [CF] = 1                      |

| Class        | Subclass | Name                           | Type | Min | Max | Default | Unit   |
|--------------|----------|--------------------------------|------|-----|-----|---------|--------|
| Fuel Gauging | CEDV Cfg | Requested Learning Cycle Count | U1   | 1   | 255 | 20      | Cycles |

### 17.15 0x0D *RelativeStateOfCharge()*

This read-word function returns the predicted remaining battery capacity as a percentage of *FullChargeCapacity()* and is an output of the CEDV gas gauging feature.

| SBS<br>Cmd | Name                           | Access |    |    | Proto-<br>col | Type | Min | Max  |
|------------|--------------------------------|--------|----|----|---------------|------|-----|------|
|            |                                | SE     | US | FA |               |      |     |      |
| 0x0D       | <i>RelativeStateOfCharge()</i> | R      |    |    | Word          | U1   | 0%  | 100% |

### 17.16 0x0E *AbsoluteStateOfCharge()*

This read-word function returns the predicted remaining battery capacity as a percentage of *Design Capacity*, and is an output of the CEDV gas gauging feature.

| SBS<br>Cmd | Name                           | Access |    |    | Proto-<br>col | Type | Min | Max  |
|------------|--------------------------------|--------|----|----|---------------|------|-----|------|
|            |                                | SE     | US | FA |               |      |     |      |
| 0x0E       | <i>AbsoluteStateOfCharge()</i> | R      |    |    | Word          | U1   | 0%  | 100% |

### 17.17 0x0F *RemainingCapacity()*

This read-word function returns the predicted remaining battery capacity and is an output of the CEDV gas gauging feature. It is scaled per *SpecificationInfo()*.

| SBS<br>Cmd | Name                       | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit          |
|------------|----------------------------|--------|----|----|---------------|------|-----|-------|---------------|
|            |                            | SE     | US | FA |               |      |     |       |               |
| 0x0F       | <i>RemainingCapacity()</i> | R      | R  | R  | Word          | U2   | 0   | 65535 | mAh<br>10 mWh |

**NOTE:** If *BatteryMode()*[CAPM] = 0, then the data reports in mAh.

If *BatteryMode()*[CAPM] = 1, then the data reports in 10 mWh.

### 17.18 0x10 *FullChargeCapacity()*

This read-word function returns the predicted battery capacity when fully charged and scaled per *SpecificationInfo()*.

| SBS<br>Cmd | Name                        | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit          |
|------------|-----------------------------|--------|----|----|---------------|------|-----|-------|---------------|
|            |                             | SE     | US | FA |               |      |     |       |               |
| 0x10       | <i>FullChargeCapacity()</i> | R      | R  | R  | Word          | U2   | 0   | 65535 | mAh<br>10 mWh |

**NOTE:** If *BatteryMode()[CAPM]* = 0, then the data reports in mAh.  
If *BatteryMode()[CAPM]* = 1, then the data reports in 10 mWh.

### 17.19 0x11 RunTimeToEmpty()

This read-word function returns the predicted remaining battery capacity based on the present rate of discharge.

| SBS<br>Cmd | Name                    | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit |
|------------|-------------------------|--------|----|----|---------------|------|-----|-------|------|
|            |                         | SE     | US | FA |               |      |     |       |      |
| 0x11       | <i>RunTimeToEmpty()</i> | R      | R  | R  | Word          | U2   | 0   | 65535 | min  |

**NOTE:** 65535 = Battery is not being discharged.

### 17.20 0x12 AverageTimeToEmpty()

This read-word function returns the predicted remaining battery capacity based on *AverageCurrent()*.

| SBS<br>Cmd | Name                        | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit |
|------------|-----------------------------|--------|----|----|---------------|------|-----|-------|------|
|            |                             | SE     | US | FA |               |      |     |       |      |
| 0x12       | <i>AverageTimeToEmpty()</i> | R      | R  | R  | Word          | U2   | 0   | 65535 | min  |

**NOTE:** 65535 = Battery is not being discharged.

### 17.21 0x13 AverageTimeToFull()

This read-word function returns the predicted time to full charge based on *AverageCurrent()*.

| SBS<br>Cmd | Name                       | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit |
|------------|----------------------------|--------|----|----|---------------|------|-----|-------|------|
|            |                            | SE     | US | FA |               |      |     |       |      |
| 0x13       | <i>AverageTimeToFull()</i> | R      | R  | R  | Word          | U2   | 0   | 65535 | min  |

**NOTE:** 65535 = Battery is not being discharged.

### 17.22 0x14 ChargingCurrent()

This read-word function returns the desired charging current.

| SBS<br>Cmd | Name                     | Access |    |    | Proto-<br>col | Type | Min | Max   | Unit |
|------------|--------------------------|--------|----|----|---------------|------|-----|-------|------|
|            |                          | SE     | US | FA |               |      |     |       |      |
| 0x14       | <i>ChargingCurrent()</i> | R      | R  | R  | Word          | U2   | 0   | 65535 | mA   |

**NOTE:** 65535 = Request maximum current  
Scaled per *SpecificationInfo()*

### 17.23 0x15 ChargingVoltage()

This read-word function returns the desired charging voltage.

| SBS Cmd | Name                     | Access |    |    | Protocol | Type | Min | Max   | Unit | Note                            |
|---------|--------------------------|--------|----|----|----------|------|-----|-------|------|---------------------------------|
|         |                          | SE     | US | FA |          |      |     |       |      |                                 |
| 0x15    | <i>ChargingVoltage()</i> | R      | R  | R  | Word     | U2   | 0   | 65535 | mV   | 65535 = request maximum voltage |

**NOTE:** 65535 = Request maximum voltage

Scaled per *SpecificationInfo()*

## 17.24 0x16 BatteryStatus()

This read-word function returns various battery status information.

| SBS Cmd | Name                   | Access |    |    | Protocol | Type | Min    | Max    |
|---------|------------------------|--------|----|----|----------|------|--------|--------|
|         |                        | SE     | US | FA |          |      |        |        |
| 0x16    | <i>BatteryStatus()</i> | R      | R  | R  | Word     | H2   | 0x0000 | 0xFFFF |

|     |     |      |     |     |      |     |     |      |     |    |    |     |     |     |     |
|-----|-----|------|-----|-----|------|-----|-----|------|-----|----|----|-----|-----|-----|-----|
| 15  | 14  | 13   | 12  | 11  | 10   | 9   | 8   | 7    | 6   | 5  | 4  | 3   | 2   | 1   | 0   |
| OCA | TCA | RSVD | OTA | TDA | RSVD | RCA | RTA | INIT | DSG | FC | FD | EC3 | EC2 | EC1 | EC0 |

**OCA (Bit 15):** Overcharged Alarm

1 = Active

0 = Inactive

**TCA (Bit 14):** Terminate Charge Alarm

1 = Active

0 = Inactive

**RSVD (Bit 13):** Reserved

**OTA (Bit 12):** Overtemperature Alarm

1 = Active

0 = Inactive

**TDA (Bit 11):** Terminate Discharge Alarm

1 = Active

0 = Inactive

**RSVD (Bit 10):** Reserved

**RCA (Bit 9):** Remaining Capacity Alarm

1 = Active

0 = Inactive

**RTA (Bit 8):** Remaining Time Alarm

1 = Active

0 = Inactive

**INIT (Bit 7):** Initialization

1 = Active

0 = Inactive

**DSG (Bit 6):** Discharging or Rest

1 = Battery is discharging or at rest.

0 = Battery is charging.

**FC (Bit 5): Fully Charged**

- 1 = Battery is fully charged.
- 0 = Battery is not fully charged.

**FD (Bit 4): Fully Discharged**

- 1 = Battery is fully discharged.
- 0 = Battery is ok.

**EC3:0 (Bits 3–0): Error Code**

- 0x0 = OK
- 0x1 = Busy
- 0x2 = Reserved Command
- 0x3 = Unsupported Command
- 0x4 = AccessDenied
- 0x5 = Overflow/Underflow
- 0x6 = BadSize
- 0x7 = UnknownError

## 17.25 0x17 CycleCount()

This read-word function returns the number of discharge cycles the battery has experienced. The value of *CycleCount()* increments when an accumulated discharge is more than **Cycle Count Percentage** of *FullChargeCapacity()* (if **[CCT]** = 1) or **Design Capacity** (if **[CCT]** = 0).

**NOTE:** A minimum of 10% of **Design Capacity** change of the accumulated discharge is required for cycle count increment. This prevents an erroneous cycle count increment due to extremely low *FullChargeCapacity()*.

| SBS Cmd | Name                | Access |     |     | Protocol | Type | Min | Max   | Unit   |
|---------|---------------------|--------|-----|-----|----------|------|-----|-------|--------|
|         |                     | SE     | US  | FA  |          |      |     |       |        |
| 0x17    | <i>CycleCount()</i> | R      | R/W | R/W | Word     | U2   | 0   | 65535 | cycles |

| Class        | Subclass | Name                   | Format           | Min | Max | Default | Unit | Description            |
|--------------|----------|------------------------|------------------|-----|-----|---------|------|------------------------|
| Fuel Gauging | Cycle    | Cycle Count Percentage | Unsigned Integer | 0   | 100 | 90      | %    | Cycle Count Percentage |

## 17.26 0x18 DesignCapacity()

This read-word function returns the theoretical pack capacity. The default value is stored in the data flash value **Design Capacity mAh** or **Design Capacity cWh**. The data should be entered in the same units as configured by *SpecificationInfo()* [**IPSCALE**].

**NOTE:** If *BatteryMode()*[**CAPM**] = 0, then the data reports in mAh.  
If *BatteryMode()*[**CAPM**] = 1, then the data reports in 10 mWh.

| SBS Cmd | Name                    | Access |     |     | Protocol | Type             | Min | Max   | Unit   |
|---------|-------------------------|--------|-----|-----|----------|------------------|-----|-------|--------|
|         |                         | SE     | US  | FA  |          |                  |     |       |        |
| 0x18    | <i>DesignCapacity()</i> | R      | R/W | R/W | Word     | Unsigned Integer | 0   | 65535 | mAh    |
|         |                         |        |     |     |          |                  |     |       | 10 mWh |



| Class        | Subclass | Name                   | Format  | Length in Bytes | Min | Max   | Default | Unit   |
|--------------|----------|------------------------|---------|-----------------|-----|-------|---------|--------|
| Fuel Gauging | Design   | Design Capacity mAh    | Integer | 2               | 0   | 32767 | 4400    | mAh    |
| Fuel Gauging | Design   | Design Capacity 10 mWh | Integer | 2               | 0   | 32767 | 6336    | 10 mWh |

## 17.27 0x19 DesignVoltage()

This read-word function returns the theoretical cell nominal voltage. The default value is stored in data flash value **Design Voltage**.

| SBS Cmd | Name            | Access |     |     | Protocol | Type             | Length in Bytes | Min | Max   | Unit |
|---------|-----------------|--------|-----|-----|----------|------------------|-----------------|-----|-------|------|
|         |                 | SE     | US  | FA  |          |                  |                 |     |       |      |
| 0x19    | DesignVoltage() | R      | R/W | R/W | Word     | Unsigned Integer | 2               | 0   | 65535 | mV   |

| Class        | Subclass | Name           | Format  | Length in Bytes | Min | Max  | Default | Unit |
|--------------|----------|----------------|---------|-----------------|-----|------|---------|------|
| Fuel Gauging | Design   | Design Voltage | Integer | 2               | 0   | 5000 | 3600    | mV   |

## 17.28 0x1A SpecificationInfo()

This read-word function returns, as an unsigned integer value, the version number of the Smart Battery Specification that the battery pack supports, as well as voltage- and current-scaling information.

Power-scaling is the product of the voltage-scaling × the current-scaling. The data is packed, as follows:

$IPScale \times 0x1000 + VScale \times 0x0100 + SpecID\_H \times 0x0010 + SpecID\_L$

**NOTE:** IPScale is only supported in the BQ78350-tbd-R2 device. For applications using higher than 32767 mA, update to the BQ78350-tbd-R2 firmware.

VScale (voltage scaling) should be selected based on the total battery pack maximum voltage. The default for this is 1, which causes the battery level voltage data to be reported in 10-mV units rather than 1-mV units. The IPScale (current scaling) should be selected based on the battery size and current magnitudes expected. The default for this is 1, which causes all mA and mAh units to change to 10 mA and 10 mAh, respectively. The default setting is stored in **Data: Specification Information**, and if the data flash value is changed, a reset (POR or via *ManufacturerAccess() [RESET]*) of the device is required for full correct operation.

The default value is stored in the following data flash.

| Class             | Subclass | Name                      | Format | Length in Bytes | Min    | Max    | Default | Unit |
|-------------------|----------|---------------------------|--------|-----------------|--------|--------|---------|------|
| SBS Configuration | Data     | Specification Information | Hex    | 2               | 0x0000 | 0xFFFF | 0x0031  | —    |

|         |    |    |    |        |    |   |   |         |   |   |   |          |   |   |   |
|---------|----|----|----|--------|----|---|---|---------|---|---|---|----------|---|---|---|
| 15      | 14 | 13 | 12 | 11     | 10 | 9 | 8 | 7       | 6 | 5 | 4 | 3        | 2 | 1 | 0 |
| IPSCALE |    |    |    | VSCALE |    |   |   | VERSION |   |   |   | REVISION |   |   |   |

### IPSCALE (Bit 15:12): Current Scale Factor

0x0 = Noted reported current and capacities are scaled by 10E0.

0x1 = Noted reported current and capacities are scaled by 10E1.

0x2 = Noted reported current and capacities are scaled by 10E2.

0x3 = Noted reported current and capacities are scaled by 10E2.

### VSCALE (Bit 11:8): Voltage Scale Factor

0x0 = *Voltage()* and *ChargingVoltage()* are scaled by 10E0.

0x1 = *Voltage()* and *ChargingVoltage()* are scaled by 10E1.

0x2 = *Voltage()* and *ChargingVoltage()* are scaled by 10E1.

0x3 = *Voltage()* and *ChargingVoltage()* are scaled by 10E1.

#### VER (Bit 7:4): Version

0x1 = Version 1.0

0x2 = Version 1.1

0x3 = Version 1.1 with optional PEC support

#### REV3:0 (Bit 3:0): Revision

0x1 = Version 1.0 and 1.1 (default)

All others reserved

### 17.29 0x1B ManufacturerDate()

This read-word function returns the pack's manufacture date. The *ManufacturerDate()* value is in the following format: Day + Month × 32 + (Year–1980) × 512, and is stored in **Manufacturer Date**.

| SBS<br>Cmd | Name                      | Access |     |     | Protocol | Type                | Length in<br>Bytes | Min | Max   | Unit |
|------------|---------------------------|--------|-----|-----|----------|---------------------|--------------------|-----|-------|------|
|            |                           | SE     | US  | FA  |          |                     |                    |     |       |      |
| 0x1B       | <i>ManufacturerDate()</i> | R      | R/W | R/W | Word     | Unsigned<br>Integer | 2                  | 0   | 65535 | —    |

| Class             | Subclass | Name              | Format              | Length in<br>Bytes | Min | Max   | Default  | Unit |
|-------------------|----------|-------------------|---------------------|--------------------|-----|-------|----------|------|
| SBS Configuration | Data     | Manufacturer Date | Unsigned<br>Integer | 2                  | 0   | 65535 | 01/01/80 |      |

### 17.30 0x1C SerialNumber()

This read-word function returns the assigned pack serial number stored in **Serial Number**.

| SBS<br>Cmd | Name                  | Access |     |     | Protocol | Type                | Length in<br>Bytes | Min | Max   | Unit |
|------------|-----------------------|--------|-----|-----|----------|---------------------|--------------------|-----|-------|------|
|            |                       | SE     | US  | FA  |          |                     |                    |     |       |      |
| 0x1C       | <i>SerialNumber()</i> | R      | R/W | R/W | Word     | Unsigned<br>Integer | 2                  | 0   | 65535 | —    |

| Class             | Subclass | Name          | Format | Length in Bytes | Min    | Max    | Default | Unit |
|-------------------|----------|---------------|--------|-----------------|--------|--------|---------|------|
| SBS Configuration | Data     | Serial Number | Hex    | 2               | 0x0000 | 0xFFFF | 0x0001  |      |

### 17.31 0x20 ManufacturerName()

This read-block function returns the pack manufacturer's name stored in **Manufacturer Name**.

| SBS<br>Cmd | Name                      | Access |    |    | Protocol | Type   | Length in Bytes | Min | Max | Unit |
|------------|---------------------------|--------|----|----|----------|--------|-----------------|-----|-----|------|
|            |                           | SE     | US | FA |          |        |                 |     |     |      |
| 0x20       | <i>ManufacturerName()</i> | R      | R  | R  | Block    | String | 20              | —   | —   | —    |

| Class             | Subclass | Name              | Format | Length in Bytes | Min | Max | Default              | Unit  |
|-------------------|----------|-------------------|--------|-----------------|-----|-----|----------------------|-------|
| SBS Configuration | Data     | Manufacturer Name | String | 20              | —   | —   | Texas<br>Instruments | ASCII |

### 17.32 0x21 DeviceName()

This read-block function returns the assigned pack name stored in **Device Name**.

| SBS Cmd | Name         | Access |    |    | Protocol | Type   | Length in Bytes | Min | Max | Unit  |
|---------|--------------|--------|----|----|----------|--------|-----------------|-----|-----|-------|
|         |              | SE     | US | FA |          |        |                 |     |     |       |
| 0x21    | DeviceName() | R      | R  | R  | Block    | String | 7+1             | —   | —   | ASCII |

| Class             | Subclass | Name        | Format | Length in Bytes | Min | Max | Default     | Unit  |
|-------------------|----------|-------------|--------|-----------------|-----|-----|-------------|-------|
| SBS Configuration | Data     | Device Name | String | 8               | —   | —   | BQ78350-R1A | ASCII |

### 17.33 0x22 DeviceChemistry()

This read-block function returns the battery chemistry used in the pack stored in **Device Chemistry**.

| SBS Cmd | Name              | Access |    |    | Protocol | Type   | Length in Bytes | Min | Max | Unit  |
|---------|-------------------|--------|----|----|----------|--------|-----------------|-----|-----|-------|
|         |                   | SE     | US | FA |          |        |                 |     |     |       |
| 0x22    | DeviceChemistry() | R      | R  | R  | Block    | String | 4+1             | —   | —   | ASCII |

| Class             | Subclass | Name             | Format | Length in Bytes | Min | Max | Default | Unit  |
|-------------------|----------|------------------|--------|-----------------|-----|-----|---------|-------|
| SBS Configuration | Data     | Device Chemistry | String | 5               | —   | —   | LION    | ASCII |

### 17.34 0x23 ManufacturerData()/CalibrationData()

This read-block function returns several elements of manufacturing-related pack information in the default mode. It is also used to return a variety of other data and can provide measured voltage, current, and temperature data for calibration purposes in CALIBRATION mode. See [Section 17.2](#), *ManufacturerAccess()*, for more details on these options.

| SBS Cmd | Name               | Access |    |    | Protocol | Type  | Min | Max | Unit | Note |
|---------|--------------------|--------|----|----|----------|-------|-----|-----|------|------|
|         |                    | SE     | US | FA |          |       |     |     |      |      |
| 0x23    | ManufacturerData() | R      | R  | R  | Block    | S32+1 |     |     |      |      |

| Status           | Condition          | Action                                                |
|------------------|--------------------|-------------------------------------------------------|
| ManufacturerData | Valid command sent | Returns pack information on <i>ManufacturerData()</i> |

The default data returned by *ManufacturerData()* is stored in the following data flash.

| Class             | Subclass | Name              | Format | Length in Bytes | Min | Max | Default | Unit  |
|-------------------|----------|-------------------|--------|-----------------|-----|-----|---------|-------|
| SBS Configuration | Data     | Manufacturer Info | String | 33              | —   | —   | —       | ASCII |

When the BQ78350-R1A is in CALIBRATION mode, this command changes the data it returns. See [Calibration](#) for more details on the data and format.

| SBS Cmd | Name              | Access |    |    | Protocol | Type   | Min | Max | Unit | Note |
|---------|-------------------|--------|----|----|----------|--------|-----|-----|------|------|
|         |                   | SE     | US | FA |          |        |     |     |      |      |
| 0x23    | CalibrationData() | R      | R  | R  | Block    | H2+S24 |     |     |      |      |

| Status          | Condition                                                                                                                                                                        | Action                                                                         |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| CalibrationData | 0x002D to <i>ManufacturerAccess()</i> to enable CALIBRATION mode<br>0xF081 or 0xF082 to <i>ManufacturerAccess()</i> to enable calibration data acquisition<br>Valid command sent | Return measured voltage, current, and temperature on <i>ManufacturerData()</i> |

### 17.35 0x2B HostFETControl

This read/write-word function enables the host to control the PCHG, DSG, and CHG FETs during normal operation (if protection features allow). The following two-step procedure enables the host to control the FETs via SMBus commands:

1. Send the FET control access code [0x1197] to *HostFETControl()*.
2. Send data to turn on or off the required FET to *HostFETControl()* within 4 s of sending the FET control access code.

The sequence must be repeated for each write to control the FETs. When the FETs are under control from the *HostFETControl()* command, then *[HCFET]* in *OperationStatus()* is set. If there are any SMBus commands received by the BQ78350-R1A in between receiving the FET control access code and the FET control data, then the FET control data is ignored. The host must wait 4 s to start another sequence to control the FETs whether the previous sequence was successful or not. The two-step sequence enables control of the FETs in SEALED mode.

| SBS<br>Cmd | Name                    | Access |     |     | Protocol | Type | Min    | Max   | Unit | Note |
|------------|-------------------------|--------|-----|-----|----------|------|--------|-------|------|------|
|            |                         | SE     | US  | FA  |          |      |        |       |      |      |
| 0x2B       | <i>HostFETControl()</i> | R/W    | R/W | R/W | Word     | Hex  | 0x0000 | 0x003 | —    |      |

|      |      |      |      |      |      |     |     |
|------|------|------|------|------|------|-----|-----|
| 7    | 6    | 5    | 4    | 3    | 2    | 1   | 0   |
| RSVD | RSVD | RSVD | RSVD | RSVD | PCHG | DSG | CHG |

**RSVD (Bits 7–3):** Reserved

**PCHG (Bit 2):** Pre-Charge FET Control

1 = ON, if protection features allow, see [XCHG]

0 = OFF

**DSG (Bit 1):** Discharge FET Control

1 = ON, if protection features allow, see [XDSG]

0 = OFF

**CHG (Bit 0):** Charge FET Control

1 = ON, if protection features allow, see [XCHG]

0 = OFF

### 17.36 0x2C GPIOStatus

This read-word function returns the bit-wise status of all the GPIO enabled in *GPIO Config*.

| SBS<br>Cmd | Name                | Access |    |    | Protocol | Type | Min    | Max    | Unit | Note |
|------------|---------------------|--------|----|----|----------|------|--------|--------|------|------|
|            |                     | SE     | US | FA |          |      |        |        |      |      |
| 0x2C       | <i>GPIOStatus()</i> | R      | R  | R  | Word     | Hex  | 0x0000 | 0x00EF | —    |      |

|        |        |      |      |      |      |      |      |
|--------|--------|------|------|------|------|------|------|
| 7      | 6      | 5    | 4    | 3    | 2    | 1    | 0    |
| GPIO B | GPIO A | LED5 | LED4 | LED3 | LED2 | LED1 | RSVD |

**GPIO B (Bit 7):** GPIO B

1 = High

0 = Low

**GPIO A (Bit 6):** GPIO A

1 = High

0 = Low

**LED5 (Bit 5):** GPIO 5

1 = High

0 = Low

**LED4 (Bit 4):** GPIO 4

1 = High

0 = Low

**LED3 (Bit 3):** GPIO 3

1 = High

0 = Low

**LED2 (Bit 2):** GPIO 2

1 = High

0 = Low

**LED1 (Bit 1):** GPIO 1

1 = High

0 = Low

**RSVD (Bit 0):** Reserved

### 17.37 0x2D GPIOControl

This read/write-word function allows the host to program GPIO Outputs enabled in **GPIO Type** to a High or Low state. A Read of this command returns what was written to the command. To read the actual status of the GPIO pins, the *GPIOStatus()* command should be used.

| SBS Cmd | Name          | Access |     |     | Protocol | Type | Min    | Max    | Unit | Note |
|---------|---------------|--------|-----|-----|----------|------|--------|--------|------|------|
|         |               | SE     | US  | FA  |          |      |        |        |      |      |
| 0x2D    | GPIOControl() | R/W    | R/W | R/W | Word     | Hex  | 0x0000 | 0x00EF | —    |      |

|        |        |      |      |      |      |      |      |
|--------|--------|------|------|------|------|------|------|
| 7      | 6      | 5    | 4    | 3    | 2    | 1    | 0    |
| GPIO B | GPIO A | LED5 | LED4 | LED3 | LED2 | LED1 | RSVD |

**GPIO B (Bit 7):** GPIO B

1 = High

0 = Low

**GPIO A (Bit 6):** GPIO A

1 = High

0 = Low

**LED5 (Bit 5):** GPIO 5

1 = High

0 = Low

**LED4 (Bit 4):** GPIO 4

1 = High

0 = Low

**LED3 (Bit 3):** GPIO 3

1 = High

0 = Low

**LED2 (Bit 2):** GPIO 2

1 = High

0 = Low

**LED1 (Bit 1):** GPIO 1

1 = High

0 = Low

**RSVD (Bit 0):** Reserved

### 17.38 0x2E VAUXVoltage()

This read-word function returns an unsigned integer value representing the scaled measured voltage from the VAUX pin, in units of mV if **[VAUX\_SCALE] = 0**, and 10 mV if **[VAUX\_SCALE] = 1** with a range of 0 to 65535. The returned value = VAUX × **VAUX Gain**.

| SBS Cmd | Name          | Access |    |    | Protocol | Type | Min | Max | Unit | Note |
|---------|---------------|--------|----|----|----------|------|-----|-----|------|------|
|         |               | SE     | US | FA |          |      |     |     |      |      |
| 0x2E    | VAUXVoltage() | R      | R  | R  | Word     | U2   | —   | —   | —    |      |

The value of **VAUX Gain** is stored in the following data flash.

| Class       | Subclass | Name      | Type | Min | Max   | Default | Unit |
|-------------|----------|-----------|------|-----|-------|---------|------|
| Calibration | Voltage  | VAUX Gain | U4   | 0   | 65535 | 5000    | —    |

### 17.39 0x2F Authenticate()/ManufacturerInput()

This read-/write-block function provides SHA-1 authentication in the default mode. It is also used to perform data flash read/writes in DATA FLASH ACCESS mode.

| SBS Cmd | Name           | Access |     |     | Protocol | Type  | Min | Max | Unit | Note |
|---------|----------------|--------|-----|-----|----------|-------|-----|-----|------|------|
|         |                | SE     | US  | FA  |          |       |     |     |      |      |
| 0x2F    | Authenticate() | R/W    | R/W | R/W | Block    | H20+1 | —   | —   | —    |      |

| SBS Cmd | Name                | Access |     |     | Protocol | Type | Min | Max | Unit | Note |
|---------|---------------------|--------|-----|-----|----------|------|-----|-----|------|------|
|         |                     | SE     | US  | FA  |          |      |     |     |      |      |
| 0x2F    | ManufacturerInput() | R/W    | R/W | R/W | Block    | H32  | —   | —   | —    |      |

### 17.40 0x30..0x3E CellVoltage1..15()

These read-word functions return an unsigned value of the calculated individual cell voltages, in mV, with a range of 0 to 65535. **CellVoltage1()** corresponds to the bottom-most series cell element, while **CellVoltage15()** corresponds to the top-most series cell element.

| SBS Cmd | Mode | Name          | Format           | Size in Bytes | Min | Max   | Default | Unit |
|---------|------|---------------|------------------|---------------|-----|-------|---------|------|
| 0x30    | R    | Reserved      | Unsigned Integer | 2             | 0   | 65535 | —       | mV   |
| 0x31    |      | CellVoltage15 |                  |               |     |       |         |      |
| 0x32    |      | CellVoltage14 |                  |               |     |       |         |      |
| 0x33    |      | CellVoltage13 |                  |               |     |       |         |      |
| 0x34    |      | CellVoltage12 |                  |               |     |       |         |      |
| 0x35    |      | CellVoltage11 |                  |               |     |       |         |      |
| 0x36    |      | CellVoltage10 |                  |               |     |       |         |      |
| 0x37    |      | CellVoltage9  |                  |               |     |       |         |      |
| 0x38    |      | CellVoltage8  |                  |               |     |       |         |      |
| 0x39    |      | CellVoltage7  |                  |               |     |       |         |      |
| 0x3A    |      | CellVoltage6  |                  |               |     |       |         |      |
| 0x3B    |      | CellVoltage5  |                  |               |     |       |         |      |
| 0x3C    |      | CellVoltage4  |                  |               |     |       |         |      |
| 0x3D    |      | CellVoltage3  |                  |               |     |       |         |      |
| 0x3E    |      | CellVoltage2  |                  |               |     |       |         |      |
| 0x3F    |      | CellVoltage1  |                  |               |     |       |         |      |

#### 17.41 0x4C *DynamicPower()*

The BQ78350-R1A includes the read-only *DynamicPower()* word command, which returns a value of *Current()* × *Voltage()* in 10 mW units. If *Current()* and/or *Voltage()* are scaled, then the units of *DynamicPower()* are also scaled.

| SBS Cmd | Mode | Name          | Format  | Size in Bytes | Min    | Max   | Default | Unit  |
|---------|------|---------------|---------|---------------|--------|-------|---------|-------|
| 0x4C    | R    | Dynamic Power | Integer | 2             | −32768 | 32767 | —       | 10 mW |

#### 17.42 0x4D *ExtAveCellVoltage()*

This read-word function returns the external average cell voltage measurement, if enabled, which can be used within the gas gauging algorithm for EDV2, EDV1, and EDV0 detection.

| SBS Cmd | Mode | Name               | Format           | Size in Bytes | Min | Max   | Default | Unit |
|---------|------|--------------------|------------------|---------------|-----|-------|---------|------|
| 0x4D    | R    | ExtAveCell Voltage | Unsigned Integer | 2             | 0   | 65535 | —       | mV   |

#### 17.43 0x4E *PendingEDV()*

This read-word function returns the predicted EDV2 until EDV2 is reached, then the predicted EDV1 until EDV1 is reached, and then the predicted EDV0. The format is Little Endian.

| SBS Cmd | Mode | Name       | Format           | Size in Bytes | Min | Max   | Default | Unit |
|---------|------|------------|------------------|---------------|-----|-------|---------|------|
| 0x4E    | R    | PendingEDV | Unsigned Integer | 2             | 0   | 65535 | —       | mV   |

## 17.44 0x4F StateOfHealth (SOH)

This command returns the state-of-health (SOH) information of the battery. *StateOfHealth()* is calculated as a percentage of *FullChargeCapacity()* Design Capacity or Design Energy. It is a read-only command.

## 17.45 0x50 SafetyAlert

This command returns the *SafetyAlert()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                 | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit | Note |
|------------|----------------------|--------|----|----|----------|------|----------------|----------------|---------|------|------|
|            |                      | SE     | US | FA |          |      |                |                |         |      |      |
| 0x50       | <i>SafetyAlert()</i> | —      | R  | R  | Block    | H4   | 0x00000<br>000 | 0xFFFF<br>FFFF | —       | —    |      |

## 17.46 0x51 SafetyStatus

This command returns the *SafetyStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                  | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit | Note |
|------------|-----------------------|--------|----|----|----------|------|----------------|----------------|---------|------|------|
|            |                       | SE     | US | FA |          |      |                |                |         |      |      |
| 0x51       | <i>SafetyStatus()</i> | —      | R  | R  | Block    | H4   | 0x00000<br>000 | 0xFFFF<br>FFFF | —       | —    |      |

## 17.47 0x52 PFAAlert

This command returns the *PFAAlert()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name              | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit | Note |
|------------|-------------------|--------|----|----|----------|------|----------------|----------------|---------|------|------|
|            |                   | SE     | US | FA |          |      |                |                |         |      |      |
| 0x52       | <i>PFAAlert()</i> | —      | R  | R  | Block    | H4   | 0x00000<br>000 | 0xFFFF<br>FFFF | —       | —    |      |

## 17.48 0x53 PFStatus

This command returns the *PFStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name              | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit | Note |
|------------|-------------------|--------|----|----|----------|------|----------------|----------------|---------|------|------|
|            |                   | SE     | US | FA |          |      |                |                |         |      |      |
| 0x53       | <i>PFStatus()</i> | —      | R  | R  | Block    | H4   | 0x00000<br>000 | 0xFFFF<br>FFFF | —       | —    |      |

## 17.49 0x54 OperationStatus

This command returns the *OperationStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                     | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit | Note |
|------------|--------------------------|--------|----|----|----------|------|----------------|----------------|---------|------|------|
|            |                          | SE     | US | FA |          |      |                |                |         |      |      |
| 0x54       | <i>OperationStatus()</i> | —      | R  | R  | Block    | H4   | 0x00000<br>000 | 0xFFFF<br>FFFF | —       | —    |      |



## 17.50 0x55 ChargingStatus

This command returns the *ChargingStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                    | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit | Note |
|------------|-------------------------|--------|----|----|----------|------|----------------|----------------|---------|------|------|
|            |                         | SE     | US | FA |          |      |                |                |         |      |      |
| 0x55       | <i>ChargingStatus()</i> | —      | R  | R  | Block    | H4   | 0x00000<br>000 | 0xFFFF<br>FFFF | —       | —    |      |

## 17.51 0x56 GaugingStatus

This command returns the *GaugingStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                   | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit | Note |
|------------|------------------------|--------|----|----|----------|------|----------------|----------------|---------|------|------|
|            |                        | SE     | US | FA |          |      |                |                |         |      |      |
| 0x56       | <i>GaugingStatus()</i> | —      | R  | R  | Block    | H4   | 0x00000<br>000 | 0xFFFF<br>FFFF | —       | —    |      |

## 17.52 0x57 ManufacturingStatus

This command returns the *ManufacturingStatus()* flags. For a description of each bit flag, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                         | Access |    |    | Protocol | Type | Min            | Max            | Default | Unit | Note |
|------------|------------------------------|--------|----|----|----------|------|----------------|----------------|---------|------|------|
|            |                              | SE     | US | FA |          |      |                |                |         |      |      |
| 0x57       | <i>ManufacturingStatus()</i> | —      | R  | R  | Block    | H4   | 0x00000<br>000 | 0xFFFF<br>FFFF | —       | —    |      |

## 17.53 0x58 AFESTatus

This command sets or gets the configuration of the companion AFE (address 0x00).

| SBS<br>Cmd | Name               | Access |    |    | Protocol | Type | Min  | Max  | Default | Unit | Note |
|------------|--------------------|--------|----|----|----------|------|------|------|---------|------|------|
|            |                    | SE     | US | FA |          |      |      |      |         |      |      |
| 0x58       | <i>AFESTatus()</i> | —      | R  | R  | Block    | H1   | 0x00 | 0xFF | —       | —    |      |

## 17.54 0x59 AFEConfig

This read/write-block function sets or gets the configuration of the companion AFE (address 0x01 to 0x0B).

| SBS<br>Cmd | Name               | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|--------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                    | SE     | US | FA |          |      |     |     |         |      |      |
| 0x59       | <i>AFEConfig()</i> | —      | R  | R  | String   | H4   | —   | —   | —       | —    |      |

## 17.55 0x5A AFEVCx

This command returns the cell voltage measurement data of the companion AFE (address 0x0C to 0x29) on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| SBS<br>Cmd | Name            | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|-----------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                 | SE     | US | FA |          |      |     |     |         |      |      |
| 0x5A       | <i>AFEVCx()</i> | —      | R  | R  | String   | H4   | —   | —   | —       | —    |      |

## 17.56 0x5B AFEData

This command returns the system voltage, temperature, and current measurement data of the companion AFE (address 0x2A to 0x33, 0x50, 0x51, and 0x59) on *ManufacturerBlockAccess()* or *ManufacturerData()*.

| SBS<br>Cmd | Name             | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                  | SE     | US | FA |          |      |     |     |         |      |      |
| 0x5B       | <i>AFEData()</i> | —      | R  | R  | String   | H4   | —   | —   | —       | —    |      |

## 17.57 0x60 Lifetime Data Block 1

This command returns the first block of Lifetime data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |      |
| 0x60       | <i>LifeTimeDataBlock1()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.58 0x61 Lifetime Data Block 2

This command returns the second block of Lifetime data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |      |
| 0x61       | <i>LifeTimeDataBlock2()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.59 0x62 Lifetime Data Block 3

This command returns the third block of Lifetime data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |      |
| 0x62       | <i>LifeTimeDataBlock3()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.60 0x63 Lifetime Data Block 4

This command returns the third block of Lifetime data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |      |
| 0x63       | <i>LifeTimeDataBlock4()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.61 0x64 Lifetime Data Block 5

This command returns the third block of Lifetime data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |      |
| 0x64       | <i>LifeTimeDataBlock5()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.62 0x65 Lifetime Data Block 6

This command returns the third block of Lifetime data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |      |
| 0x65       | <i>LifeTimeDataBlock6()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.63 0x66 Lifetime Data Block 7

This command returns the third block of Lifetime data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                        | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|-----------------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                             | SE     | US | FA |          |      |     |     |         |      |      |
| 0x66       | <i>LifeTimeDataBlock7()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.64 0x70 ManufacturerInfo

This command returns manufacturer information. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                      | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|---------------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                           | SE     | US | FA |          |      |     |     |         |      |      |
| 0x70       | <i>ManufacturerInfo()</i> | R      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.65 0x71 DAStatus1

This command returns the Cell Voltages. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name               | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|--------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                    | SE     | US | FA |          |      |     |     |         |      |      |
| 0x71       | <i>DAStatus1()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.66 0x72 DAStatus2

This command returns the TS1, TS2, TS3, and cell temperatures. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name               | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|--------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                    | SE     | US | FA |          |      |     |     |         |      |      |
| 0x72       | <i>DAStatus2()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.67 0x80 CUV Snapshot

This command returns the CUV snapshot data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name                 | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|----------------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |                      | SE     | US | FA |          |      |     |     |         |      |      |
| 0x80       | <i>CUVSnapshot()</i> | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## 17.68 0x81 COV Snapshot

This command returns the COV snapshot data. For a description of returned data values, see the *ManufacturerAccess()* version of the same command in [Section 17.2](#).

| SBS<br>Cmd | Name          | Access |    |    | Protocol | Type | Min | Max | Default | Unit | Note |
|------------|---------------|--------|----|----|----------|------|-----|-----|---------|------|------|
|            |               | SE     | US | FA |          |      |     |     |         |      |      |
| 0x81       | COVSnapshot() | —      | R  | R  | Block    | —    | —   | —   | —       | —    |      |

## Data Flash Access and Format

### 18.1 Data Flash Access

#### 18.1.1 Minimum Voltage

Data flash can only be updated when the sum of the cell voltages connected between the VC0 and VC5 pins of the companion AFE is above the **Valid Update Voltage**.

| Class | Subclass | Name                 | Type | Min | Max   | Default | Unit | Description                            |
|-------|----------|----------------------|------|-----|-------|---------|------|----------------------------------------|
| Power | Power    | Valid Update Voltage | I2   | 0   | 32767 | 3500    | mV   | Min voltage threshold for Flash update |

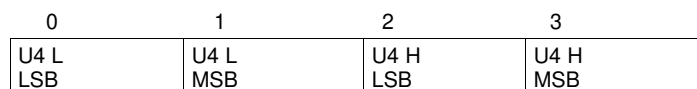
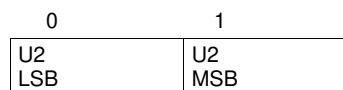
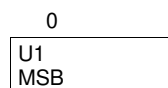
**NOTE:**

- VC1 to VC5 voltages may not be represented directly by *CellVoltage1...5()* when the number of cells in series is > 5. The VCx to *CellVoltage*x() decode is through the DF:AFE Cell Map.
- It is not recommended to change this value.

### 18.2 Data Formats

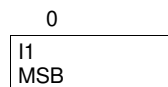
#### 18.2.1 Unsigned Integer (U)

Unsigned integer is stored without changes as 1-byte, 2-byte, or 4-byte values in Little Endian byte order.



#### 18.2.2 Integer (I)

Integer values are stored in 2's-complement format in 1-byte, 2-byte, or 4-byte values in Little Endian byte order.



| 0         | 1         |
|-----------|-----------|
| I2<br>LSB | I2<br>MSB |

| 0           | 1           | 2           | 3           |
|-------------|-------------|-------------|-------------|
| I4 L<br>LSB | I4 L<br>MSB | I4 H<br>LSB | I4 H<br>MSB |

### 18.2.3 Floating Point (F)

Floating point values are stored using a 4-byte format, where the LSB is the exponent, bytes 1 to 3 are the mantissa in unsigned integer format, with the MSB in byte 1 as a signed bit.

| 0        | 1                     | 2           | 3          |
|----------|-----------------------|-------------|------------|
| Exp[0–7] | Sign + Fract<br>[0–6] | Fract[7–14] | Exp[15–22] |

Where:

- $s = 128$  for negative numbers, 0 otherwise.
- $\text{Fract}[0-6] < 127$

### 18.2.4 Hex (H)

Bit register definitions are stored in unsigned integer format.

### 18.2.5 String (S)

String values are stored with length byte first, followed by a number of data bytes defined with the length byte.

| 0      | 1     | ...   | N     |
|--------|-------|-------|-------|
| Length | Data0 | . . . | DataN |

## Data Flash Summary

**Table 19-1. Data Flash Summary Table**

| Class       | Subclass               | Address | Name                           | Type | Min Value   | Max Value    | Default    | Units |
|-------------|------------------------|---------|--------------------------------|------|-------------|--------------|------------|-------|
| Calibration | Current                | 0x4000  | CC Gain                        | F4   | 1.00E–01    | 9.00E+00     | 8.4381     | —     |
| Calibration | Current                | 0x4004  | Capacity Gain                  | F4   | 2.98262E+04 | 2.693046E+07 | 2516761.36 | —     |
| Calibration | Voltage                | 0x4008  | Cell1 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x4009  | Cell2 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x400A  | Cell3 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x400B  | Cell4 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x400C  | Cell5 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x400D  | Cell6 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x400E  | Cell7 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x400F  | Cell8 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x4010  | Cell9 Offset                   | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x4011  | Cell10 Offset                  | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x4012  | Cell11 Offset                  | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x4013  | Cell12 Offset                  | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x4014  | Cell13 Offset                  | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x4015  | Cell14 Offset                  | I1   | –128        | 127          | 0          | mV    |
| Calibration | Voltage                | 0x4016  | Cell15 Offset                  | I1   | –128        | 127          | 0          | mV    |
| Calibration | Current Offset         | 0x4018  | CC Offset                      | I2   | –32767      | 32767        | 0          | —     |
| Calibration | Current Offset         | 0x401A  | Coulomb Counter Offset Samples | U2   | 0           | 65535        | 64         | —     |
| Calibration | Temperature            | 0x401E  | T1 Temp Offset                 | I1   | –128        | 127          | 0          | 0.1°C |
| Calibration | Temperature            | 0x401F  | T2 Temp Offset                 | I1   | –128        | 127          | 0          | 0.1°C |
| Calibration | Temperature            | 0x4020  | T3 Temp Offset                 | I1   | –128        | 127          | 0          | 0.1°C |
| Calibration | Temperature            | 0x4021  | Gauge Internal Temp Offset     | I1   | –128        | 127          | 0          | 0.1°C |
| Calibration | Ext Cell Voltage       | 0x4022  | Ext Cell Divider Gain          | I2   | 0           | 32767        | 5000       | —     |
| Calibration | VAux Voltage           | 0x4024  | VAux Gain                      | I4   | 0           | 100000       | 5000       | —     |
| Calibration | Cell Temperature Model | 0x454F  | Coeff a1                       | I2   | –32768      | 32767        | –11130     | —     |
| Calibration | Cell Temperature Model | 0x4551  | Coeff a2                       | I2   | –32768      | 32767        | 19142      | —     |
| Calibration | Cell Temperature Model | 0x4553  | Coeff a3                       | I2   | –32768      | 32767        | –19262     | —     |
| Calibration | Cell Temperature Model | 0x4555  | Coeff a4                       | I2   | –32768      | 32767        | 28203      | —     |
| Calibration | Cell Temperature Model | 0x4557  | Coeff a5                       | I2   | –32768      | 32767        | 892        | —     |
| Calibration | Cell Temperature Model | 0x4559  | Coeff b1                       | I2   | –32768      | 32767        | 328        | —     |

**Table 19-1. Data Flash Summary Table (continued)**

| Class       | Subclass               | Address | Name                       | Type | Min Value | Max Value | Default | Units  |
|-------------|------------------------|---------|----------------------------|------|-----------|-----------|---------|--------|
| Calibration | Cell Temperature Model | 0x455B  | Coeff b2                   | I2   | –32768    | 32767     | –605    | —      |
| Calibration | Cell Temperature Model | 0x455D  | Coeff b3                   | I2   | –32768    | 32767     | –2443   | —      |
| Calibration | Cell Temperature Model | 0x455F  | Coeff b4                   | I2   | –32768    | 32767     | 4696    | —      |
| Calibration | Cell Temperature Model | 0x4561  | Rc0                        | I2   | –32768    | 32767     | 11703   | —      |
| Calibration | Cell Temperature Model | 0x4563  | Adc0                       | I2   | –32768    | 32767     | 11703   | —      |
| Calibration | Fet Temperature Model  | 0x4569  | Coeff a1                   | I2   | –32768    | 32767     | –11130  | —      |
| Calibration | Fet Temperature Model  | 0x456B  | Coeff a2                   | I2   | –32768    | 32767     | 19142   | —      |
| Calibration | Fet Temperature Model  | 0x456D  | Coeff a3                   | I2   | –32768    | 32767     | –19262  | —      |
| Calibration | Fet Temperature Model  | 0x456F  | Coeff a4                   | I2   | –32768    | 32767     | 28203   | —      |
| Calibration | Fet Temperature Model  | 0x4571  | Coeff a5                   | I2   | –32768    | 32767     | 892     | —      |
| Calibration | Fet Temperature Model  | 0x4573  | Coeff b1                   | I2   | –32768    | 32767     | 328     | —      |
| Calibration | Fet Temperature Model  | 0x4575  | Coeff b2                   | I2   | –32768    | 32767     | –605    | —      |
| Calibration | Fet Temperature Model  | 0x4577  | Coeff b3                   | I2   | –32768    | 32767     | –2443   | —      |
| Calibration | Fet Temperature Model  | 0x4579  | Coeff b4                   | I2   | –32768    | 32767     | 4696    | —      |
| Calibration | Fet Temperature Model  | 0x457B  | Rc0                        | I2   | –32768    | 32767     | 11703   | —      |
| Calibration | Fet Temperature Model  | 0x457D  | Adc0                       | I2   | –32768    | 32767     | 11703   | —      |
| Calibration | Fet Temperature Model  | 0x457F  | Rpad                       | I2   | –32768    | 32767     | 0       | —      |
| Calibration | Fet Temperature Model  | 0x4581  | Rint                       | I2   | –32768    | 32767     | 18000   | —      |
| Calibration | Current Deadband       | 0x4318  | Deadband                   | U1   | 0         | 255       | 3       | mA     |
| Calibration | Current Deadband       | 0x4319  | Coulomb Counter Deadband   | U1   | 0         | 255       | 38      | 264 nV |
| Settings    | Protection             | 0x44C5  | Protection Configuration   | H1   | 0x0       | 0x3E      | 0x00    | Hex    |
| Settings    | Protection             | 0x44C6  | Enabled Protections A      | H1   | 0x0       | 0xFF      | 0xFF    | Hex    |
| Settings    | Protection             | 0x44C7  | Enabled Protections B      | H1   | 0x0       | 0x7F      | 0x7F    | Hex    |
| Settings    | Protection             | 0x44C8  | Enabled Protections C      | H1   | 0x0       | 0x1F      | 0x15    | Hex    |
| Settings    | Protection             | 0x44CA  | Enabled Removal Recovery A | H1   | 0x0       | 0xFC      | 0x0     | Hex    |



**Table 19-1. Data Flash Summary Table (continued)**

| Class       | Subclass          | Address | Name                       | Type | Min Value | Max Value | Default | Units |
|-------------|-------------------|---------|----------------------------|------|-----------|-----------|---------|-------|
| Settings    | Protection        | 0x44CB  | Enabled Removal Recovery B | H1   | 0x0       | 0x40      | 0x0     | Hex   |
| Settings    | Permanent Failure | 0x451D  | Enabled PF A               | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Settings    | Permanent Failure | 0x451E  | Enabled PF B               | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Settings    | Configuration     | 0x445F  | FET Options                | H2   | 0x0       | 0x03FF    | 0x0021  | Hex   |
| Settings    | Configuration     | 0x4461  | Sbs Gauging Configuration  | H1   | 0x0       | 0x83      | 0x0     | Hex   |
| Settings    | Configuration     | 0x4462  | Smb Configuration          | H1   | 0x0       | 0xB7      | 0xA0    | Hex   |
| Settings    | Configuration     | 0x4472  | Power Config               | H1   | 0x0       | 0x01      | 0x00    | Hex   |
| Settings    | Configuration     | 0x448C  | LED Configuration          | H1   | 0x0       | 0xFF      | 0x00    | Hex   |
| Settings    | Configuration     | 0x44A9  | Temperature Enable         | H1   | 0x0       | 0x1F      | 0x09    | Hex   |
| Settings    | Configuration     | 0x44AA  | Temperature Mode           | H1   | 0x0       | 0x07      | 0x0     | Hex   |
| Settings    | Configuration     | 0x44AB  | DA Configuration           | H1   | 0x0       | 0xDF      | 0x11    | Hex   |
| Settings    | Configuration     | 0x44AC  | AFE Cell Map               | H2   | 0x0       | 0x7FFF    | 0x0013  | Hex   |
| Settings    | Configuration     | 0x458B  | CEDV Gauging Configuration | H2   | 0x0       | 0x07FF    | 0x0002  | Hex   |
| Settings    | Configuration     | 0x45D4  | SOC Flag Config            | H2   | 0x0       | 0x13FF    | 0x02FB  | Hex   |
| Settings    | Configuration     | 0x460C  | Balancing Configuration    | H1   | 0x0       | 0xFF      | 0x01    | Hex   |
| Settings    | Fuse              | 0x4458  | PF SAFE A                  | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Settings    | Fuse              | 0x4459  | PF SAFE B                  | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Settings    | Fuse              | 0x445A  | PF SAFE C                  | H1   | 0x0       | 0x03      | 0x0     | Hex   |
| Settings    | Fuse              | 0x445E  | Fuse Blow Timeout          | U1   | 0         | 255       | 30      | s     |
| Settings    | Aux SMB Address   | 0x4463  | Addr Reads                 | U1   | 0         | 10        | 3       | —     |
| Settings    | Aux SMB Address   | 0x4464  | SMBTAR_ADDR0               | H1   | 0x0       | 0xFF      | 0x20    | Hex   |
| Settings    | Aux SMB Address   | 0x4465  | SMBTAR_ADDR1               | H1   | 0x0       | 0xFF      | 0x22    | Hex   |
| Settings    | Aux SMB Address   | 0x4466  | SMBTAR_ADDR2               | H1   | 0x0       | 0xFF      | 0x24    | Hex   |
| Settings    | Aux SMB Address   | 0x4467  | SMBTAR_ADDR3               | H1   | 0x0       | 0xFF      | 0x25    | Hex   |
| Settings    | Aux SMB Address   | 0x4468  | SMBTAR_ADDR4               | H1   | 0x0       | 0xFF      | 0x26    | Hex   |
| Settings    | Aux SMB Address   | 0x4469  | SMBTAR_ADDR5               | H1   | 0x0       | 0xFF      | 0x28    | Hex   |
| Settings    | Aux SMB Address   | 0x446A  | SMBTAR_ADDR6               | H1   | 0x0       | 0xFF      | 0x2A    | Hex   |
| Settings    | Aux SMB Address   | 0x446B  | SMBTAR_ADDR7               | H1   | 0x0       | 0xFF      | 0x2C    | Hex   |
| Settings    | SMB Master Mode   | 0x446E  | Host Address               | H1   | 0x0       | 0xFF      | 0x10    | Hex   |
| Settings    | SMB Master Mode   | 0x446F  | Charger Address            | H1   | 0x0       | 0xFF      | 0x12    | Hex   |
| Settings    | SMB Master Mode   | 0x4470  | Alarm Timer                | U1   | 10        | 255       | 10      | s     |
| Settings    | SMB Master Mode   | 0x4471  | Charger Request Timer      | U1   | 10        | 255       | 50      | s     |
| Settings    | Manufacturing     | 0x42C8  | Mfg Status init            | H2   | 0x0       | 0xFFFF    | 0x0000  | Hex   |
| Protections | CUV               | 0x44CC  | Threshold                  | I2   | 0         | 32767     | 2500    | mV    |
| Protections | CUV               | 0x44CE  | Delay                      | U1   | 0         | 255       | 2       | s     |
| Protections | CUV               | 0x44CF  | Recovery                   | I2   | 0         | 32767     | 3000    | mV    |
| Protections | COV               | 0x44D1  | Threshold                  | I2   | 0         | 32767     | 4300    | mV    |
| Protections | COV               | 0x44D3  | Delay                      | U1   | 0         | 255       | 2       | s     |
| Protections | COV               | 0x44D4  | Recovery                   | I2   | 0         | 32767     | 4100    | mV    |

**Table 19-1. Data Flash Summary Table (continued)**

| Class       | Subclass              | Address | Name                | Type | Min Value | Max Value | Default | Units |
|-------------|-----------------------|---------|---------------------|------|-----------|-----------|---------|-------|
| Protections | OCC                   | 0x44D6  | Threshold           | I2   | –32768    | 32767     | 6000    | mA    |
| Protections | OCC                   | 0x44D8  | Delay               | U1   | 0         | 255       | 6       | s     |
| Protections | OCC                   | 0x44D9  | Recovery Threshold  | I2   | –32768    | 32767     | –200    | mA    |
| Protections | OCC                   | 0x44DB  | Recovery Delay      | U1   | 0         | 255       | 5       | s     |
| Protections | OCD                   | 0x44DC  | Threshold           | I2   | –32768    | 32767     | –6000   | mA    |
| Protections | OCD                   | 0x44DE  | Delay               | U1   | 0         | 255       | 6       | s     |
| Protections | OCD                   | 0x44DF  | Recovery Threshold  | I2   | –32768    | 32767     | 200     | mA    |
| Protections | OCD                   | 0x44E1  | Recovery Delay      | U1   | 0         | 255       | 5       | s     |
| Protections | OCDL                  | 0x44E2  | Latch Limit         | U1   | 0         | 255       | 5       | —     |
| Protections | OCDL                  | 0x44E3  | Counter Dec Delay   | U1   | 0         | 255       | 10      | s     |
| Protections | OCDL                  | 0x44E4  | Reset               | U1   | 0         | 255       | 15      | s     |
| Protections | AOLD                  | 0x44B1  | Threshold and Delay | H1   | 0x0       | 0xFF      | 0x00    | Hex   |
| Protections | AOLD                  | 0x44E5  | Latch Limit         | U1   | 0         | 255       | 0       | —     |
| Protections | AOLD                  | 0x44E6  | Counter Dec Delay   | U1   | 0         | 255       | 10      | s     |
| Protections | AOLD                  | 0x44E7  | Recovery            | U1   | 0         | 255       | 5       | s     |
| Protections | AOLD                  | 0x44E8  | Reset               | U1   | 0         | 255       | 15      | s     |
| Protections | ASCD                  | 0x44B0  | Threshold and Delay | H1   | 0x0       | 0xFF      | 0x00    | Hex   |
| Protections | ASCD                  | 0x44E9  | Latch Limit         | U1   | 0         | 255       | 0       | —     |
| Protections | ASCD                  | 0x44EA  | Counter Dec Delay   | U1   | 0         | 255       | 10      | s     |
| Protections | ASCD                  | 0x44EB  | Recovery            | U1   | 0         | 255       | 5       | s     |
| Protections | ASCD                  | 0x44EC  | Reset               | U1   | 0         | 255       | 15      | s     |
| Protections | OTC                   | 0x44ED  | Threshold           | I2   | –400      | 1500      | 550     | 0.1°C |
| Protections | OTC                   | 0x44EF  | Delay               | U1   | 0         | 255       | 2       | s     |
| Protections | OTC                   | 0x44F0  | Recovery            | I2   | –400      | 1500      | 500     | 0.1°C |
| Protections | OTD                   | 0x44F2  | Threshold           | I2   | –400      | 1500      | 600     | 0.1°C |
| Protections | OTD                   | 0x44F4  | Delay               | U1   | 0         | 255       | 2       | s     |
| Protections | OTD                   | 0x44F5  | Recovery            | I2   | –400      | 1500      | 550     | 0.1°C |
| Protections | OTF                   | 0x44F7  | Threshold           | I2   | –400      | 1500      | 800     | 0.1°C |
| Protections | OTF                   | 0x44F9  | Delay               | U1   | 0         | 255       | 2       | s     |
| Protections | OTF                   | 0x44FA  | Recovery            | I2   | –400      | 1500      | 650     | 0.1°C |
| Protections | UTC                   | 0x44FC  | Threshold           | I2   | –400      | 1500      | 0       | 0.1°C |
| Protections | UTC                   | 0x44FE  | Delay               | U1   | 0         | 255       | 2       | s     |
| Protections | UTC                   | 0x44FF  | Recovery            | I2   | –400      | 1500      | 50      | 0.1°C |
| Protections | UTD                   | 0x4501  | Threshold           | I2   | –400      | 1500      | 0       | 0.1°C |
| Protections | UTD                   | 0x4503  | Delay               | U1   | 0         | 255       | 2       | s     |
| Protections | UTD                   | 0x4504  | Recovery            | I2   | –400      | 1500      | 50      | 0.1°C |
| Protections | AFE External Override | 0x4506  | Delay               | U1   | 0         | 255       | 5       | s     |
| Protections | AFE External Override | 0x4507  | Recovery            | U1   | 0         | 255       | 2       | s     |
| Protections | PTO                   | 0x4508  | Charge Threshold    | I2   | –32768    | 32767     | 2000    | mA    |
| Protections | PTO                   | 0x450A  | Suspend Threshold   | I2   | –32768    | 32767     | 1800    | mA    |
| Protections | PTO                   | 0x450C  | Delay               | U2   | 0         | 65535     | 1800    | s     |
| Protections | PTO                   | 0x450E  | Reset               | I2   | 0         | 32767     | 2       | mAh   |
| Protections | CTO                   | 0x4510  | Charge Threshold    | I2   | –32768    | 32767     | 2500    | mA    |
| Protections | CTO                   | 0x4512  | Suspend Threshold   | I2   | –32768    | 32767     | 2000    | mA    |
| Protections | CTO                   | 0x4514  | Delay               | U2   | 0         | 65535     | 54000   | s     |
| Protections | CTO                   | 0x4516  | Reset               | I2   | 0         | 32767     | 2       | mAh   |
| Protections | OC                    | 0x4518  | Threshold           | I2   | –32768    | 32767     | 300     | mAh   |
| Protections | OC                    | 0x451A  | Recovery            | I2   | –32768    | 32767     | 2       | mAh   |
| Protections | OC                    | 0x451C  | RSOC Recovery       | U1   | 0         | 100       | 90      | %     |

**Table 19-1. Data Flash Summary Table (continued)**

| Class            | Subclass              | Address | Name                            | Type | Min Value | Max Value | Default | Units |
|------------------|-----------------------|---------|---------------------------------|------|-----------|-----------|---------|-------|
| Permanent Fail   | SOC                   | 0x4521  | Threshold                       | I2   | –32768    | 32767     | 10000   | mA    |
| Permanent Fail   | SOC                   | 0x4523  | Delay                           | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | SOC                   | 0x4524  | Threshold                       | I2   | –32768    | 32767     | –10000  | mA    |
| Permanent Fail   | SOC                   | 0x4526  | Delay                           | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | SOT                   | 0x4527  | Threshold                       | I2   | –400      | 1500      | 650     | 0.1°C |
| Permanent Fail   | SOT                   | 0x4529  | Delay                           | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | SOTF                  | 0x452A  | Threshold                       | I2   | –400      | 1500      | 1000    | 0.1°C |
| Permanent Fail   | SOTF                  | 0x452C  | Delay                           | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | Open Thermistor       | 0x452D  | Threshold                       | I2   | 0         | 32767     | 2232    | 0.1 K |
| Permanent Fail   | Open Thermistor       | 0x452F  | Delay                           | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | VIMR                  | 0x4530  | Check Voltage                   | I2   | 0         | 5000      | 3500    | mV    |
| Permanent Fail   | VIMR                  | 0x4532  | Check Current                   | I2   | 0         | 32767     | 10      | mA    |
| Permanent Fail   | VIMR                  | 0x4534  | Delta Threshold                 | I2   | 0         | 5000      | 500     | mV    |
| Permanent Fail   | VIMR                  | 0x4536  | Delta Delay                     | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | VIMR                  | 0x4537  | Duration                        | U2   | 0         | 65535     | 100     | s     |
| Permanent Fail   | CFET                  | 0x4539  | OFF Threshold                   | I2   | 0         | 500       | 5       | mA    |
| Permanent Fail   | CFET                  | 0x453B  | OFF Delay                       | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | DFET                  | 0x453C  | OFF Threshold                   | I2   | –500      | 0         | –5      | mA    |
| Permanent Fail   | DFET                  | 0x453E  | OFF Delay                       | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | AFER                  | 0x453F  | Threshold                       | U1   | 0         | 255       | 100     | —     |
| Permanent Fail   | AFER                  | 0x4540  | Delay Period                    | U1   | 0         | 255       | 2       | s     |
| Permanent Fail   | AFER                  | 0x4541  | Compare Period                  | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | AFEC                  | 0x4542  | Threshold                       | U1   | 0         | 255       | 100     | —     |
| Permanent Fail   | AFEC                  | 0x4543  | Delay Period                    | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | AFE XREADY            | 0x4544  | Threshold                       | U1   | 0         | 255       | 100     | —     |
| Permanent Fail   | AFE XREADY            | 0x4545  | Delay Period                    | U1   | 0         | 255       | 5       | s     |
| Permanent Fail   | AFE External Override | 0x4546  | Delay                           | U1   | 0         | 255       | 20      | s     |
| Permanent Fail   | AFE SOV/AFE SUV       | 0x44B2  | SOV and SUV Delay               | H1   | 0x0       | 0xFF      | 0x50    | Hex   |
| Permanent Fail   | AFE SOV               | 0x44B7  | Threshold                       | I2   | 3150      | 4700      | 4350    | mV    |
| Permanent Fail   | AFE SUV               | 0x44B9  | Threshold                       | I2   | 1580      | 3100      | 1750    | mV    |
| Charge Algorithm | Temperature Ranges    | 0x45DF  | Charge Inhibit/Suspend Low Temp | I1   | –128      | 127       | 0       | °C    |

**Table 19-1. Data Flash Summary Table (continued)**

| Class            | Subclass              | Address | Name                         | Type | Min Value | Max Value | Default | Units |
|------------------|-----------------------|---------|------------------------------|------|-----------|-----------|---------|-------|
| Charge Algorithm | Temperature Ranges    | 0x45E0  | Precharge Temp               | I1   | –128      | 127       | 12      | °C    |
| Charge Algorithm | Temperature Ranges    | 0x45E1  | Charge Inhibit High Temp     | I1   | –128      | 127       | 45      | °C    |
| Charge Algorithm | Temperature Ranges    | 0x45E2  | Charge Suspend High Temp     | I1   | –128      | 127       | 55      | °C    |
| Charge Algorithm | Temperature Ranges    | 0x45E3  | Hysteresis Temp              | I1   | –128      | 127       | 3       | °C    |
| Charge Algorithm | Fast Charging         | 0x45E4  | Voltage                      | I2   | 0         | 32767     | 4200    | mV    |
| Charge Algorithm | Fast Charging         | 0x45E6  | Current                      | I2   | 0         | 32767     | 3000    | mA    |
| Charge Algorithm | Pre-Charging          | 0x45E8  | Current                      | I2   | 0         | 32767     | 100     | mA    |
| Charge Algorithm | Pre-Charging          | 0x45EA  | Start Voltage                | I2   | 0         | 32767     | 2500    | mV    |
| Charge Algorithm | Pre-Charging          | 0x45EC  | Recovery Voltage             | I2   | 0         | 32767     | 2900    | mV    |
| Charge Algorithm | Termination Config    | 0x45EE  | Charge Term Taper Current    | I2   | 0         | 32767     | 250     | mA    |
| Charge Algorithm | Termination Config    | 0x45F2  | Charge Term Voltage          | I2   | 0         | 32767     | 75      | mV    |
| Charge Algorithm | Cell Balancing Config | 0x460D  | Cell Balance Threshold       | I2   | 0         | 5000      | 3900    | mV    |
| Charge Algorithm | Cell Balancing Config | 0x460F  | Cell Balance Window          | I2   | 0         | 5000      | 100     | mV    |
| Charge Algorithm | Cell Balancing Config | 0x4611  | Cell Balance Min             | U1   | 0         | 255       | 40      | mV    |
| Charge Algorithm | Cell Balancing Config | 0x4612  | Cell Balance Interval        | U1   | 0         | 255       | 20      | s     |
| Fuel Gauging     | Current Thresholds    | 0x44BD  | Dsg Current Threshold        | I2   | –32768    | 32767     | 100     | mA    |
| Fuel Gauging     | Current Thresholds    | 0x44BF  | Chg Current Threshold        | I2   | –32768    | 32767     | 50      | mA    |
| Fuel Gauging     | Current Thresholds    | 0x44C1  | Quit Current                 | I2   | 0         | 32767     | 10      | mA    |
| Fuel Gauging     | Current Thresholds    | 0x44C3  | Dsg Relax Time               | U1   | 0         | 255       | 1       | s     |
| Fuel Gauging     | Current Thresholds    | 0x44C4  | Chg Relax Time               | U1   | 0         | 255       | 60      | s     |
| Fuel Gauging     | Design                | 0x45CC  | Design Capacity mAh          | I2   | 0         | 32767     | 4400    | mAh   |
| Fuel Gauging     | Design                | 0x45CE  | Design Capacity cWh          | I2   | 0         | 32767     | 6336    | cWh   |
| Fuel Gauging     | Design                | 0x45D0  | Design Voltage               | I2   | 0         | 5000      | 3600    | mV    |
| Fuel Gauging     | Cycle                 | 0x45D2  | Cycle Count Percentage       | U1   | 0         | 100       | 90      | %     |
| Fuel Gauging     | FD                    | 0x45D6  | Set % RSOC Threshold         | U1   | 0         | 100       | 0       | %     |
| Fuel Gauging     | FD                    | 0x45D7  | Clear % RSOC Threshold       | U1   | 0         | 100       | 5       | %     |
| Fuel Gauging     | FC                    | 0x45D8  | Set % RSOC Threshold         | U1   | 0         | 100       | 100     | %     |
| Fuel Gauging     | FC                    | 0x45D9  | Clear % RSOC Threshold       | U1   | 0         | 100       | 95      | %     |
| Fuel Gauging     | TD                    | 0x45DA  | Set % RSOC Threshold         | U1   | 0         | 100       | 6       | %     |
| Fuel Gauging     | TD                    | 0x45DB  | Clear % RSOC Threshold       | U1   | 0         | 100       | 8       | %     |
| Fuel Gauging     | TC                    | 0x45DC  | Set % RSOC Threshold         | U1   | 0         | 100       | 100     | %     |
| Fuel Gauging     | TC                    | 0x45DD  | Clear % RSOC Threshold       | U1   | 0         | 100       | 95      | %     |
| Fuel Gauging     | State                 | 0x42C2  | Learned Full Charge Capacity | I2   | 0         | 32767     | 4400    | mAh   |
| Fuel Gauging     | State                 | 0x42C4  | Dod at EDV2                  | I2   | 0         | 16384     | 15232   | —     |
| Fuel Gauging     | State                 | 0x42C6  | Cycle Count                  | U2   | 0         | 65535     | 0       | —     |
| Fuel Gauging     | CEDV cfg              | 0x458D  | EMF                          | U2   | 0         | 65535     | 3743    | —     |
| Fuel Gauging     | CEDV cfg              | 0x458F  | C0                           | U2   | 0         | 65535     | 149     | —     |

**Table 19-1. Data Flash Summary Table (continued)**

| Class        | Subclass           | Address | Name                           | Type | Min Value | Max Value | Default | Units     |
|--------------|--------------------|---------|--------------------------------|------|-----------|-----------|---------|-----------|
| Fuel Gauging | CEDV cfg           | 0x4591  | R0                             | U2   | 0         | 65535     | 867     | —         |
| Fuel Gauging | CEDV cfg           | 0x4593  | T0                             | U2   | 0         | 65535     | 4030    | —         |
| Fuel Gauging | CEDV cfg           | 0x4595  | R1                             | U2   | 0         | 65535     | 316     | —         |
| Fuel Gauging | CEDV cfg           | 0x4597  | TC                             | U1   | 0         | 255       | 9       | —         |
| Fuel Gauging | CEDV cfg           | 0x4598  | C1                             | U1   | 0         | 255       | 0       | —         |
| Fuel Gauging | CEDV cfg           | 0x4599  | Age Factor                     | U1   | 0         | 255       | 0       | —         |
| Fuel Gauging | CEDV cfg           | 0x459A  | Fixed EDV 0                    | I2   | 0         | 32767     | 3031    | —         |
| Fuel Gauging | CEDV cfg           | 0x459C  | EDV 0 Hold Time                | U1   | 1         | 255       | 1       | s         |
| Fuel Gauging | CEDV cfg           | 0x459D  | Fixed EDV 1                    | I2   | 0         | 32767     | 3385    | —         |
| Fuel Gauging | CEDV cfg           | 0x459F  | EDV 1 Hold Time                | U1   | 1         | 255       | 1       | s         |
| Fuel Gauging | CEDV cfg           | 0x45A0  | Fixed EDV 2                    | I2   | 0         | 32767     | 3501    | —         |
| Fuel Gauging | CEDV cfg           | 0x45A2  | EDV 2 Hold Time                | U1   | 1         | 255       | 1       | s         |
| Fuel Gauging | CEDV cfg           | 0x45A7  | Battery Low %                  | U2   | 0         | 65535     | 700     | .01%      |
| Fuel Gauging | CEDV cfg           | 0x45AB  | Min Delta V Filter             | I2   | 0         | 32767     | 10      | mV        |
| Fuel Gauging | CEDV cfg           | 0x45AD  | FCC Learn Up                   | I2   | 0         | 32767     | 512     | mAh       |
| Fuel Gauging | CEDV cfg           | 0x45AF  | FCC Learn Down                 | I2   | 0         | 32767     | 256     | mAh       |
| Fuel Gauging | CEDV cfg           | 0x45B4  | Learning Low Temp              | I2   | –100      | 32767     | 119     | 0.1°C     |
| Fuel Gauging | CEDV cfg           | 0x45BD  | Requested Learning cycle count | U1   | 0         | 255       | 20      | Num       |
| Fuel Gauging | CEDV cfg           | 0x45BE  | OverLoad Current               | I2   | 0         | 32767     | 5000    | mA        |
| Fuel Gauging | CEDV cfg           | 0x45C2  | Self Discharge Rate            | U1   | 0         | 255       | 20      | 0.01%/day |
| Fuel Gauging | CEDV cfg           | 0x45C3  | Electronics Load               | I2   | 0         | 255       | 0       | 3 $\mu$ A |
| Fuel Gauging | CEDV cfg           | 0x45C5  | Near Full                      | I2   | 0         | 32767     | 200     | mAh       |
| Fuel Gauging | CEDV cfg           | 0x45C7  | Reserve Capacity               | I2   | 0         | 32767     | 0       | mAh       |
| Fuel Gauging | CEDV cfg           | 0x45CB  | RemCap Init Percent            | U1   | 0         | 110       | 100     | %         |
| Power        | Power              | 0x4473  | Valid Update Voltage           | I2   | 0         | 32767     | 3500    | mV        |
| Power        | Shutdown           | 0x4475  | Shutdown Voltage               | I2   | 0         | 32767     | 1750    | mV        |
| Power        | Shutdown           | 0x4477  | Shutdown Time                  | U1   | 0         | 255       | 10      | s         |
| Power        | Shutdown           | 0x4478  | PF Shutdown Voltage            | I2   | 0         | 32767     | 1750    | mV        |
| Power        | Shutdown           | 0x447A  | PF Shutdown Time               | U1   | 0         | 255       | 10      | s         |
| Power        | Shutdown           | 0x447B  | Charger Present Threshold      | I2   | 0         | 32767     | 3000    | mV        |
| Power        | Sleep              | 0x447D  | Sleep Current                  | I2   | 0         | 32767     | 10      | mA        |
| Power        | Sleep              | 0x447F  | Bus Timeout                    | U1   | 0         | 255       | 5       | s         |
| Power        | Sleep              | 0x4484  | Voltage Time                   | U1   | 0         | 255       | 5       | s         |
| Power        | Sleep              | 0x4485  | Current Time                   | U1   | 0         | 255       | 20      | s         |
| Power        | Ship               | 0x4487  | FET Off Time                   | U1   | 0         | 127       | 10      | s         |
| Power        | Ship               | 0x4488  | Delay                          | U1   | 0         | 254       | 20      | s         |
| Power        | Ship               | 0x4489  | Auto Ship Time                 | U2   | 0         | 65535     | 1440    | min       |
| Power        | KEYIN              | 0x448B  | Time                           | U1   | 0         | 10        | 2       | s         |
| PF Status    | Device Status Data | 0x41C0  | Safety Alert A                 | H1   | 0x0       | 0xFF      | 0x0     | Hex       |
| PF Status    | Device Status Data | 0x41C1  | Safety Status A                | H1   | 0x0       | 0xFF      | 0x0     | Hex       |
| PF Status    | Device Status Data | 0x41C2  | Safety Alert B                 | H1   | 0x0       | 0xFF      | 0x0     | Hex       |
| PF Status    | Device Status Data | 0x41C3  | Safety Status B                | H1   | 0x0       | 0xFF      | 0x0     | Hex       |
| PF Status    | Device Status Data | 0x41C4  | Safety Alert C                 | H1   | 0x0       | 0xFF      | 0x0     | Hex       |
| PF Status    | Device Status Data | 0x41C5  | Safety Status C                | H1   | 0x0       | 0xFF      | 0x0     | Hex       |
| PF Status    | Device Status Data | 0x41C8  | PF Alert A                     | H1   | 0x0       | 0xFF      | 0x0     | Hex       |

**Table 19-1. Data Flash Summary Table (continued)**

| Class     | Subclass                | Address | Name                   | Type | Min Value | Max Value | Default | Units |
|-----------|-------------------------|---------|------------------------|------|-----------|-----------|---------|-------|
| PF Status | Device Status Data      | 0x41C9  | PF Status A            | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41CA  | PF Alert B             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41CB  | PF Status B            | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41CC  | PF Alert C             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41CD  | PF Status C            | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41D0  | Fuse Flag              | H2   | 0x0       | 0xFFFF    | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41D2  | Operation Status A     | H2   | 0x0       | 0xFFFF    | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41D4  | Operation Status B     | H2   | 0x0       | 0xFFFF    | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41D6  | Temp Range             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41D7  | Charging Status        | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41D8  | Gauging Status         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Status Data      | 0x41D9  | CEDV Status            | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status | Device Voltage Data     | 0x41DA  | Cell 1 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41DC  | Cell 2 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41DE  | Cell 3 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41E0  | Cell 4 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41E2  | Cell 5 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41E4  | Cell 6 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41E6  | Cell 7 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41E8  | Cell 8 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41EA  | Cell 9 Voltage         | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41EC  | Cell 10 Voltage        | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41EE  | Cell 11 Voltage        | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41F0  | Cell 12 Voltage        | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41F2  | Cell 13 Voltage        | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41F4  | Cell 14 Voltage        | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41F6  | Cell 15 Voltage        | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Voltage Data     | 0x41FA  | Battery Direct Voltage | I2   | –32768    | 32767     | 0       | mV    |
| PF Status | Device Current Data     | 0x41FE  | Current                | I2   | –32768    | 32767     | 0       | mA    |
| PF Status | Device Temperature Data | 0x4200  | TS1 Temperature        | I2   | –32768    | 32767     | 0       | 0.1 K |

**Table 19-1. Data Flash Summary Table (continued)**

| Class       | Subclass                | Address | Name                        | Type | Min Value | Max Value | Default | Units |
|-------------|-------------------------|---------|-----------------------------|------|-----------|-----------|---------|-------|
| PF Status   | Device Temperature Data | 0x4202  | TS2 Temperature             | I2   | –32768    | 32767     | 0       | 0.1 K |
| PF Status   | Device Temperature Data | 0x4204  | TS3 Temperature             | I2   | –32768    | 32767     | 0       | 0.1 K |
| PF Status   | Device Temperature Data | 0x4206  | Gauge Internal Temperature  | I2   | –32768    | 32767     | 0       | 0.1 K |
| PF Status   | AFE Regs                | 0x4208  | AFE SYS Stat                | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x4209  | AFE Cell Balance1           | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x420A  | AFE Cell Balance2           | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x420B  | AFE Cell Balance3           | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x420C  | AFE Sys Control1            | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x420D  | AFE Sys Control2            | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x420E  | AFE Protection1             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x420F  | AFE Protection2             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x4210  | AFE Protection3             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x4211  | AFE OV Trip                 | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| PF Status   | AFE Regs                | 0x4212  | AFE UV Trip                 | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x4180  | 1st Status Status A         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x4181  | 1st Status Status B         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x4182  | 1st Safety Status C         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x4184  | 1st Time to Next Event      | U1   | 0         | 255       | 0       | s     |
| Black Box   | Safety Status           | 0x4185  | 2nd Status Status A         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x4186  | 2nd Status Status B         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x4187  | 2nd Safety Status C         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x4189  | 2nd Time to Next Event      | U1   | 0         | 255       | 0       | s     |
| Black Box   | Safety Status           | 0x418A  | 3rd Status Status A         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x418B  | 3rd Status Status B         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x418C  | 3rd Safety Status C         | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | Safety Status           | 0x418E  | 3rd Time to Next Event      | U1   | 0         | 255       | 0       | s     |
| Black Box   | PF Status               | 0x418F  | 1st PF Status A             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x4190  | 1st PF Status B             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x4191  | 1st PF Status C             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x4193  | 1st Time to Next Event      | U1   | 0         | 255       | 0       | s     |
| Black Box   | PF Status               | 0x4194  | 2nd PF Status A             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x4195  | 2nd PF Status B             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x4196  | 2nd PF Status C             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x4198  | 2nd Time to Next Event      | U1   | 0         | 255       | 0       | s     |
| Black Box   | PF Status               | 0x4199  | 3rd PF Status A             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x419A  | 3rd PF Status B             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x419B  | 3rd PF Status C             | H1   | 0x0       | 0xFF      | 0x0     | Hex   |
| Black Box   | PF Status               | 0x419D  | 3rd Time to Next Event      | U1   | 0         | 255       | 0       | s     |
| System Data | Manufacturer Data       | 0x4040  | Manufacturer Info A Length  | U1   | 1         | 32        | 32      | —     |
| System Data | Manufacturer Data       | 0x4041  | Manufacturer Info Block A01 | H1   | 0x0       | 0xFF      | 0x61    | Hex   |
| System Data | Manufacturer Data       | 0x4042  | Manufacturer Info Block A02 | H1   | 0x0       | 0xFF      | 0x62    | Hex   |
| System Data | Manufacturer Data       | 0x4043  | Manufacturer Info Block A03 | H1   | 0x0       | 0xFF      | 0x63    | Hex   |
| System Data | Manufacturer Data       | 0x4044  | Manufacturer Info Block A04 | H1   | 0x0       | 0xFF      | 0x64    | Hex   |

**Table 19-1. Data Flash Summary Table (continued)**

| Class             | Subclass          | Address | Name                        | Type | Min Value | Max Value | Default           | Units |
|-------------------|-------------------|---------|-----------------------------|------|-----------|-----------|-------------------|-------|
| System Data       | Manufacturer Data | 0x4045  | Manufacturer Info Block A05 | H1   | 0x0       | 0xFF      | 0x65              | Hex   |
| System Data       | Manufacturer Data | 0x4046  | Manufacturer Info Block A06 | H1   | 0x0       | 0xFF      | 0x66              | Hex   |
| System Data       | Manufacturer Data | 0x4047  | Manufacturer Info Block A07 | H1   | 0x0       | 0xFF      | 0x67              | Hex   |
| System Data       | Manufacturer Data | 0x4048  | Manufacturer Info Block A08 | H1   | 0x0       | 0xFF      | 0x68              | Hex   |
| System Data       | Manufacturer Data | 0x4049  | Manufacturer Info Block A09 | H1   | 0x0       | 0xFF      | 0x69              | Hex   |
| System Data       | Manufacturer Data | 0x404A  | Manufacturer Info Block A10 | H1   | 0x0       | 0xFF      | 0x6A              | Hex   |
| System Data       | Manufacturer Data | 0x404B  | Manufacturer Info Block A11 | H1   | 0x0       | 0xFF      | 0x6B              | Hex   |
| System Data       | Manufacturer Data | 0x404C  | Manufacturer Info Block A12 | H1   | 0x0       | 0xFF      | 0x6C              | Hex   |
| System Data       | Manufacturer Data | 0x404D  | Manufacturer Info Block A13 | H1   | 0x0       | 0xFF      | 0x6D              | Hex   |
| System Data       | Manufacturer Data | 0x404E  | Manufacturer Info Block A14 | H1   | 0x0       | 0xFF      | 0x6E              | Hex   |
| System Data       | Manufacturer Data | 0x404F  | Manufacturer Info Block A15 | H1   | 0x0       | 0xFF      | 0x6F              | Hex   |
| System Data       | Manufacturer Data | 0x4050  | Manufacturer Info Block A16 | H1   | 0x0       | 0xFF      | 0x70              | Hex   |
| System Data       | Manufacturer Data | 0x4051  | Manufacturer Info Block A17 | H1   | 0x0       | 0xFF      | 0x71              | Hex   |
| System Data       | Manufacturer Data | 0x4052  | Manufacturer Info Block A18 | H1   | 0x0       | 0xFF      | 0x72              | Hex   |
| System Data       | Manufacturer Data | 0x4053  | Manufacturer Info Block A19 | H1   | 0x0       | 0xFF      | 0x73              | Hex   |
| System Data       | Manufacturer Data | 0x4054  | Manufacturer Info Block A20 | H1   | 0x0       | 0xFF      | 0x74              | Hex   |
| System Data       | Manufacturer Data | 0x4055  | Manufacturer Info Block A21 | H1   | 0x0       | 0xFF      | 0x75              | Hex   |
| System Data       | Manufacturer Data | 0x4056  | Manufacturer Info Block A22 | H1   | 0x0       | 0xFF      | 0x76              | Hex   |
| System Data       | Manufacturer Data | 0x4057  | Manufacturer Info Block A23 | H1   | 0x0       | 0xFF      | 0x77              | Hex   |
| System Data       | Manufacturer Data | 0x4058  | Manufacturer Info Block A24 | H1   | 0x0       | 0xFF      | 0x7A              | Hex   |
| System Data       | Manufacturer Data | 0x4059  | Manufacturer Info Block A25 | H1   | 0x0       | 0xFF      | 0x78              | Hex   |
| System Data       | Manufacturer Data | 0x405A  | Manufacturer Info Block A26 | H1   | 0x0       | 0xFF      | 0x79              | Hex   |
| System Data       | Manufacturer Data | 0x405B  | Manufacturer Info Block A27 | H1   | 0x0       | 0xFF      | 0x30              | Hex   |
| System Data       | Manufacturer Data | 0x405C  | Manufacturer Info Block A28 | H1   | 0x0       | 0xFF      | 0x31              | Hex   |
| System Data       | Manufacturer Data | 0x405D  | Manufacturer Info Block A29 | H1   | 0x0       | 0xFF      | 0x32              | Hex   |
| System Data       | Manufacturer Data | 0x405E  | Manufacturer Info Block A30 | H1   | 0x0       | 0xFF      | 0x33              | Hex   |
| System Data       | Manufacturer Data | 0x405F  | Manufacturer Info Block A31 | H1   | 0x0       | 0xFF      | 0x34              | Hex   |
| System Data       | Manufacturer Data | 0x4060  | Manufacturer Info Block A32 | H1   | 0x0       | 0xFF      | 0x35              | Hex   |
| SBS Configuration | Data              | 0x406B  | Manufacture Date            | U2   | 0         | 65535     | 0                 | date  |
| SBS Configuration | Data              | 0x406D  | Serial Number               | H2   | 0x0       | 0xFFFF    | 0x0001            | Hex   |
| SBS Configuration | Data              | 0x406F  | Manufacturer Name           | S21  | x         | x         | Texas Instruments | —     |



**Table 19-1. Data Flash Summary Table (continued)**

| Class             | Subclass   | Address | Name                      | Type | Min Value | Max Value | Default | Units       |
|-------------------|------------|---------|---------------------------|------|-----------|-----------|---------|-------------|
| SBS Configuration | Data       | 0x4084  | Device Name               | S21  | x         | x         | BQ78350 | —           |
| SBS Configuration | Data       | 0x4099  | Device Chemistry          | S5   | x         | x         | LION    | —           |
| SBS Configuration | Data       | 0x449F  | Remaining AH Cap. Alarm   | I2   | 0         | 32767     | 300     | mAh         |
| SBS Configuration | Data       | 0x44A1  | Remaining WH Cap. Alarm   | I2   | 0         | 32767     | 432     | cWh         |
| SBS Configuration | Data       | 0x44A3  | Remaining Time Alarm      | U2   | 0         | 65535     | 10      | min         |
| SBS Configuration | Data       | 0x44A5  | Initial Battery Mode      | H2   | 0x0       | 0xFFFF    | 0x0081  | Hex         |
| SBS Configuration | Data       | 0x44A7  | Specification Information | H2   | 0x0       | 0xFFFF    | 0x0031  | Hex         |
| LED Support       | LED Config | 0x448D  | LED Flash Period          | U2   | 32        | 65535     | 512     | 488 $\mu$ s |
| LED Support       | LED Config | 0x448F  | LED Blink Period          | U2   | 32        | 65535     | 1024    | 488 $\mu$ s |
| LED Support       | LED Config | 0x4491  | LED Delay                 | U2   | 16        | 65535     | 100     | 488 $\mu$ s |
| LED Support       | LED Config | 0x4493  | LED Hold Time             | U1   | 1         | 63        | 16      | 0.25 s      |
| LED Support       | LED Config | 0x4494  | LED Flash Alarm           | I1   | 0         | 100       | 10      | %           |
| LED Support       | LED Config | 0x4495  | LED Thresh 1              | I1   | 0         | 100       | 0       | %           |
| LED Support       | LED Config | 0x4496  | LED Thresh 2              | I1   | 0         | 100       | 20      | %           |
| LED Support       | LED Config | 0x4497  | LED Thresh 3              | I1   | 0         | 100       | 40      | %           |
| LED Support       | LED Config | 0x4498  | LED Thresh 4              | I1   | 0         | 100       | 60      | %           |
| LED Support       | LED Config | 0x4499  | LED Thresh 5              | I1   | 0         | 100       | 80      | %           |
| LED Support       | LED Config | 0x449A  | LCD Refresh Rate          | U1   | 20        | 100       | 35      | Hz          |
| Lifetimes         | Voltage    | 0x40C0  | Cell 1 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40C2  | Cell 2 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40C4  | Cell 3 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40C6  | Cell 4 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40C8  | Cell 5 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40CA  | Cell 6 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40CC  | Cell 7 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40CE  | Cell 8 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40D0  | Cell 9 Max Voltage        | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40D2  | Cell 10 Max Voltage       | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40D4  | Cell 11 Max Voltage       | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40D6  | Cell 12 Max Voltage       | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40D8  | Cell 13 Max Voltage       | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40DA  | Cell 14 Max Voltage       | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40DC  | Cell 15 Max Voltage       | I2   | 0         | 32767     | 0       | mV          |
| Lifetimes         | Voltage    | 0x40E0  | Cell 1 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40E2  | Cell 2 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40E4  | Cell 3 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40E6  | Cell 4 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40E8  | Cell 5 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40EA  | Cell 6 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40EC  | Cell 7 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40EE  | Cell 8 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40F0  | Cell 9 Min Voltage        | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40F2  | Cell 10 Min Voltage       | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40F4  | Cell 11 Min Voltage       | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40F6  | Cell 12 Min Voltage       | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40F8  | Cell 13 Min Voltage       | I2   | 0         | 32767     | 32767   | mV          |
| Lifetimes         | Voltage    | 0x40FA  | Cell 14 Min Voltage       | I2   | 0         | 32767     | 32767   | mV          |

**Table 19-1. Data Flash Summary Table (continued)**

| Class     | Subclass          | Address | Name                   | Type | Min Value | Max Value | Default | Units  |
|-----------|-------------------|---------|------------------------|------|-----------|-----------|---------|--------|
| Lifetimes | Voltage           | 0x40FC  | Cell 15 Min Voltage    | I2   | 0         | 32767     | 32767   | mV     |
| Lifetimes | Voltage           | 0x4100  | Max Delta Cell Voltage | I2   | 0         | 32767     | 0       | mV     |
| Lifetimes | Current           | 0x4102  | Max Charge Current     | I2   | 0         | 32767     | 0       | mA     |
| Lifetimes | Current           | 0x4104  | Max Discharge Current  | I2   | –32768    | 0         | 0       | mA     |
| Lifetimes | Current           | 0x4106  | Max Avg Dsg Current    | I2   | –32768    | 0         | 0       | mA     |
| Lifetimes | Current           | 0x4108  | Max Avg Dsg Power      | I2   | –32768    | 0         | 0       | cW     |
| Lifetimes | Temperature       | 0x410A  | Max Temp Cell          | I1   | –128      | 127       | –128    | °C     |
| Lifetimes | Temperature       | 0x410B  | Min Temp Cell          | I1   | –128      | 127       | 127     | °C     |
| Lifetimes | Temperature       | 0x410C  | Max Delta Cell Temp    | I1   | –128      | 127       | 0       | °C     |
| Lifetimes | Temperature       | 0x410D  | Max Temp Fet           | I1   | –128      | 127       | –128    | °C     |
| Lifetimes | Gauge Temperature | 0x410E  | Max Temp Int Sensor    | I1   | –128      | 127       | –128    | °C     |
| Lifetimes | Gauge Temperature | 0x410F  | Min Temp Int Sensor    | I1   | –128      | 127       | 127     | °C     |
| Lifetimes | Safety Events     | 0x4110  | No Of COV Events       | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x4112  | Last COV Event         | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Safety Events     | 0x4114  | No Of CUV Events       | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x4116  | Last CUV Event         | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Safety Events     | 0x4118  | No Of OCD Events       | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x411A  | Last OCD Event         | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Safety Events     | 0x411C  | No Of OCC Events       | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x411E  | Last OCC Event         | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Safety Events     | 0x4120  | No Of AOLD Events      | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x4122  | Last AOLD Event        | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Safety Events     | 0x4124  | No Of ASCD Events      | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x4126  | Last ASCD Event        | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Safety Events     | 0x4128  | No Of OTC Events       | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x412A  | Last OTC Event         | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Safety Events     | 0x412C  | No Of OTD Events       | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x412E  | Last OTD Event         | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Safety Events     | 0x4130  | No Of OTF Events       | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Safety Events     | 0x4132  | Last OTF Event         | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Charging Events   | 0x4134  | No Valid Charge Term   | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Charging Events   | 0x4136  | Last Valid Charge Term | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Gauging Events    | 0x4138  | No of FCC Updates      | U2   | 0         | 32767     | 0       | events |
| Lifetimes | Gauging Events    | 0x413A  | Last FCC Update        | U2   | 0         | 32767     | 0       | cycles |
| Lifetimes | Power Events      | 0x413C  | No Of Shutdowns        | U1   | 0         | 255       | 0       | events |
| Lifetimes | Cell Balancing    | 0x4140  | Cb Time Cell 1         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4141  | Cb Time Cell 2         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4142  | Cb Time Cell 3         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4143  | Cb Time Cell 4         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4144  | Cb Time Cell 5         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4145  | Cb Time Cell 6         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4146  | Cb Time Cell 7         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4147  | Cb Time Cell 8         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4148  | Cb Time Cell 9         | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x4149  | Cb Time Cell 10        | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x414A  | Cb Time Cell 11        | U1   | 0         | 255       | 0       | 2 h    |
| Lifetimes | Cell Balancing    | 0x414B  | Cb Time Cell 12        | U1   | 0         | 255       | 0       | 2 h    |

**Table 19-1. Data Flash Summary Table (continued)**

| Class     | Subclass       | Address | Name                       | Type | Min Value | Max Value | Default | Units |
|-----------|----------------|---------|----------------------------|------|-----------|-----------|---------|-------|
| Lifetimes | Cell Balancing | 0x414C  | Cb Time Cell 13            | U1   | 0         | 255       | 0       | 2 h   |
| Lifetimes | Cell Balancing | 0x414D  | Cb Time Cell 14            | U1   | 0         | 255       | 0       | 2 h   |
| Lifetimes | Cell Balancing | 0x414E  | Cb Time Cell 15            | U1   | 0         | 255       | 0       | 2 h   |
| Lifetimes | Time           | 0x4150  | Total Fw Runtime           | U2   | 0         | 65535     | 0       | 2 h   |
| Lifetimes | Time           | 0x4152  | Time Spent In UT           | U2   | 0         | 65535     | 0       | 2 h   |
| Lifetimes | Time           | 0x4154  | Time Spent In LT           | U2   | 0         | 65535     | 0       | 2 h   |
| Lifetimes | Time           | 0x4156  | Time Spent In ST           | U2   | 0         | 65535     | 0       | 2 h   |
| Lifetimes | Time           | 0x4158  | Time Spent In HT           | U2   | 0         | 65535     | 0       | 2 h   |
| Lifetimes | Time           | 0x415A  | Time Spent In OT           | U2   | 0         | 65535     | 0       | 2 h   |
| Lifetimes | Time           | 0x415C  | Time Since Last Charge     | U2   | 0         | 65535     | 0       | 2 h   |
| GPIO      | GPIO Config    | 0x449B  | GPIO Configuration         | H1   | 0x0       | 0xFF      | 0x00    | Hex   |
| GPIO      | GPIO Config    | 0x449C  | GPIO Output Enable         | H1   | 0x0       | 0xFF      | 0x00    | Hex   |
| GPIO      | GPIO Config    | 0x449D  | GPIO Default Output Enable | H1   | 0x0       | 0xFF      | 0x00    | Hex   |
| GPIO      | GPIO Config    | 0x449E  | GPIO Type                  | H1   | 0x0       | 0xFF      | 0xC0    | Hex   |

## Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Date          | Revision | Notes           |
|---------------|----------|-----------------|
| December 2019 | *        | Initial Release |

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